

S A P

SAP ABAP

Interview Questions

1. **What is an ABAP data dictionary?**- ABAP 4 data dictionary describes the logical structures of the objects used in application development and shows how they are mapped to the underlying relational database in tables/views.
2. **What are domains and data element?**- Domains: Domain is the central object for describing the technical characteristics of an attribute of an business objects. It describes the value range of the field. Data Element: It is used to describe the semantic definition of the table fields like description the field. Data element describes how a field can be displayed to end-user.
3. **What is foreign key relationship?**- A relationship which can be defined between tables and must be explicitly defined at field level. Foreign keys are used to ensure the consistency of data. Data entered should be checked against existing data to ensure that there are now contradiction. While defining foreign key relationship cardinality has to be specified. Cardinality mentions how many dependent records or how referenced records are possible.
4. **Describe data classes.**- Master data: It is the data which is seldomly changed. Transaction data: It is the data which is often changed. Organization data: It is a customizing data which is entered in the system when the system is configured and is then rarely changed. System data: It is the data which R/3 system needs for itself.
5. **What are indexes?**- Indexes are described as a copy of a database table reduced to specific fields. This data exists in sorted form. This sorting form ease fast access to the field of the tables. In order that other fields are also read, a pointer to the associated record of the actual table are included in the index. The indexes are activated along with the table and are created automatically with it in the database.
6. **Difference between transparent tables and pooled tables.**- Transparent tables: Transparent tables in the dictionary has a one-to-one relation with the table in database. Its structure corresponds to single database field. Table in the database has the same name as in the dictionary. Transparent table holds application data. Pooled tables. Pooled tables in the dictionary has a many-to-one relation with the table in database. Table in the database has the different name as in the dictionary. Pooled table are stored in table pool at the database level.
7. **What is an ABAP/4 Query?**- ABAP/4 Query is a powerful tool to generate simple reports without any coding. ABAP/4 Query can generate the following 3 simple reports: Basic List: It is the simple reports. Statistics: Reports with statistical functions like Average, Percentages. Ranked Lists: For analytical reports. - For creating a ABAP/4 Query, programmer has to create user group and a functional group. Functional group can be created using with or without logical database table. Finally, assign user group to functional group. Finally, create a query on the functional group generated.
8. **What is BDC programming?**- Transferring of large/external/legacy data into SAP system using Batch Input programming. Batch input is a automatic procedure referred to as BDC(Batch Data Communications). The central component of the transfer is a queue file which receives the data via a batch input programs and groups associated data into "sessions".
9. **What are the functional modules used in sequence in BDC?**- These are the 3 functional modules which are used in a sequence to perform a data transfer successfully using BDC programming: BDC_OPEN_GROUP - Parameters like Name of the client, sessions and user name are specified in this functional modules. BDC_INSERT - It is used to insert the data for one transaction into a session. BDC_CLOSE_GROUP - This is used to close the batch input session.
10. **What are internal tables?**- Internal tables are a standard data type object which exists only during the runtime of the program. They are used to perform table calculations on subsets of database tables and for re-organising the contents of database tables according to users need.
11. **What is ITS? What are the merits of ITS?**- ITS is a Internet Transaction Server. ITS forms an interface between HTTP server and R/3 system, which converts screen provided data by the R/3 system into HTML documents and vice-versa. Merits of ITS: A complete web transaction can be developed and tested in R/3 system. All transaction components, including those used by the ITS outside the R/3 system at runtime, can be stored in the R/3 system. The advantage of automatic language processing in the R/3 system can be utilized to language-dependent HTML documents at runtime.

12. **What is DynPro?**- DynPro is a Dynamic Programming which is a combination of screen and the associated flow logic Screen is also called as DynPro.
13. **What are screen painter and menu painter?**- Screen painter: Screen painter is a tool to design and maintain screen and its elements. It allows user to create GUI screens for the transactions. Attributes, layout, field attributes and flow logic are the elements of Screen painter. Menu painter: Menu painter is a tool to design the interface components. Status, menu bars, menu lists, F-key settings, functions and titles are the components of Menu painters. Screen painter and menu painter both are the graphical interface of an ABAP/4 applications.
14. **What are the components of SAP scripts?**- SAP scripts is a word processing tool of SAP which has the following components: Standard text. It is like a standard normal documents. Layout sets. - Layout set consists of the following components: Windows and pages, Paragraph formats, Character formats. Creating forms in the R/3 system. Every layout set consists of Header, paragraph, and character string. ABAP/4 program.
15. **What is ALV programming in ABAP? When is this grid used in ABAP?**- ALV is Application List viewer. Sap provides a set of ALV (ABAP LIST VIEWER) function modules which can be put into use to embellish the output of a report. This set of ALV functions is used to enhance the readability and functionality of any report output. Cases arise in sap when the output of a report contains columns extending more than 255 characters in length. In such cases, this set of ALV functions can help choose selected columns and arrange the different columns from a report output and also save different variants for report display. This is a very efficient tool for dynamically sorting and arranging the columns from a report output. The report output can contain up to 90 columns in the display with the wide array of display options.
16. **What are the events in ABAP/4 language?**- Initialization, At selection-screen, Start-of-selection, end-of-selection, top-of-page, end-of-page, At line-selection, At user-command, At PF, Get, At New, At LAST, AT END, AT FIRST.
17. **What is CTS and what do you know about it?**- The Change and Transport System (CTS) is a tool that helps you to organize development projects in the ABAP Workbench and in Customizing, and then transport the changes between the SAP Systems and clients in your system landscape. This documentation provides you with an overview of how to manage changes with the CTS and essential information on setting up your system and client landscape and deciding on a transport strategy. Read and follow this documentation when planning your development project.
18. **What are logical databases? What are the advantages/ dis-advantages of logical databases?**- To read data from a database tables we use logical database. A logical database provides read-only access to a group of related tables to an ABAP/4 program. Advantages: i)check functions which check that user input is complete, correct,and plausible. ii)Meaningful data selection. iii)central authorization checks for database accesses. iv)good read access performance while retaining the hierarchical data view determined by the application logic. dis advantages: i)If you donot specify a logical database in the program attributes,the GET events never occur. ii)There is no ENDGET command,so the code block associated with an event ends with the next event statement (such as another GET or an END-OF-SELECTION).
19. **What is a batch input session?**- BATCH INPUT SESSION is an intermediate step between internal table and database table. Data along with the action is stored in session ie data for screen fields, to which screen it is passed, program name behind it, and how next screen is processed.
20. **How to upload data using CATT ?**- These are the steps to be followed to Upload data through CATT: Creation of the CATT test case & recording the sample data input. Download of the source file template. Modification of the source file. Upload of the data from the source file.
21. **What is Smart Forms?**- Smart Forms allows you to create forms using a graphical design tool with robust functionality, color, and more. Additionally, all new forms developed at SAP will be created with the new Smart Form solution.
22. **How can I make a differentiation between dependent and independent data?**- Client dependent or independent transfer requirements include client specific or cross client objects in the change requests.

Workbench objects like SAPscripts are client specific, some entries in customizing are client independent. If you display the object list for one change request, and then for each object the object attributes, you will find the flag client specific. If one object in the task list has this flag on, then that transport will be client dependent.

23. **What is the difference between macro and subroutine?**- Macros can only be used in the program they are defined in and only after the definition are expanded at compilation / generation. Subroutines (FORM) can be called from both the program they are defined in and other programs. A MACRO is more or less an abbreviation for some lines of code that are used more than once or twice. A FORM is a local subroutine (which can be called external). A FUNCTION is (more or less) a subroutine that is called external. Since debugging a MACRO is not really possible, prevent the use of them (I've never used them, but seen them in action). If the subroutine is used only local (called internal) use a FORM. If the subroutine is called external (used by more than one program) use a FUNCTION.

24. **How we format the data before write statement in report ?**

We can format the reports output by using the loop events like:

1. at first
 2. at new
 3. at last
- etc check docu

What is the difference between Table and Template?

table is a dynamic and template is a static

When do we use End-of-selection?

End-of-selection event are mostly used? when we are writing HR-ABAP code. In the HR-ABAP code, data is retrieved in the Start-of-selection event and Printing on the list and all will be? done in End-of-selection event.

In events start-of-selection is default event. When we have to use this event explicitly? Why?

The default event in the ABAP is Start-of-selection. We have to call explicitly this event when you are writing other than this event? , that is when you write AT? SELECTION-SCREEN EVENTS OR INITIALIZATION EVENT etc, you have to explicitly mention the Start-of-selection event while you are writing the logic. Before these events called, all the code you have written come into this default Start-of-selection screen event.

What is the differences between ABAP and OOABAP. In which situation we use OOABAP?

OOABAP is used to develop BSP/PCUI applications and also anything involved object oriented like BADIs, SmartForms..etc. where as ABAP is used to develop traditional programs in R/3.

What is table buffer? Which type of tables used this buffer?

buffer is nothing but a memory area. table is buffered means that table information is available on application server. when you call data from database table it will come from application server. transparent and pooled tables are buffered. cluster tables can not be buffered.

What is the use of pretty printer ?

Exactly where can we link the functional module to abap coding.

Pretty Printer is used to format the ABAP Code we write in ABAP Editor, like KEY WORDS in Capitals and remaining are in small letters which is also depend on system settings.

We can call the function module in the ABAP Code. Press the Pattern button on Appl. tool bar then u will get box where u write the function module NAME which u want to call in the code by selecting the radio button CALL FUNCTION. In this way we link function module to ABAP Code.

What is the difference between SAP memory and ABAP memory?

Answer1:

data sending between main sessions using get parameter and set parameter is sap memory

data sending between internal sessions using import or export parameters is abap memory

Answer2:

sap memory is a global memory whereas abap memory is local memory.

For example, we have four programs in abap memory and assigned some variables to a particular program in abap memory then those variables can't be used by any other program in abap memory i.e., the variables are only for that program and also local to that memory, whereas sap memory can access all the abap memory or else it can perform any kind of modifications.

Answer3:

SAP memory is available to the user during the entire terminal session.

ABAP memory is available to the user during? life time? of? external session.

What is the difference between Type and Like?

Answer1:

TYPE, you assign datatype directly to the data object while declaring.

LIKE, you assign the datatype of another object to the declaring data object. The datatype is referenced indirectly.

Answer2:

Type is a keyword used to refer to a data type whereas Like is a keyword used to copy the existing properties of already existing data object.

Answer3:

type refers the existing data type

like refers the existing data object

What is Tcode SE16. For what is it used. Explain briefly?

Answer1:

SE16 is a T-code for object browser.

Generally used to search the fields of SAP Tables . and respective data.

Answer2:

se16 is a data browse and it is used to view the contents of the table and we cannot change or append new fields to the existing structure of the table as we cannot view the structure level display using the se16

What are different ABAP/4 editors? What are the differences?

The 2 editors are se38 and se80 both have the abap editor in place. In se38 you can go create programs and view online reports and basically do all the development of objects in this editor. In se80 (object navigator) there are additional features such as creating packages, module pool , function group , classes, programs (where you can create ur programs) and BSP applications .

What is difference between dialog program and a report?

Report is a executable program

Dialog is a module pool program. It has to be executed via a transaction only.

Dialog programming is used for customization of screens

How do you connect to the remote server if you are working from the office for the client in remote place.

WAS web application server or ITS are generally used for this purpose. If you are sitting at your office with a server which is in the system and the other server is at the client's place you can generate IDOC, intermediate documents which carry the data you want to transfer or the documents you want to transfer, these IDOC are interpreted by the system at the receiving end with the message class with which it is bound with. If you want to logon a system which is very distant..then remote login can be used this depends on the internet speed.

Explain about roll area , Dispatcher, ABAP-Processor.

Answer1:

Roll area is nothing but memory allocated by work process. It holds the information needed by R/3 about programs execution such as value of the variables.

Dispatcher :All the requests that come from presentation server will be directed first to dispatcher. Further dispatcher sends this requests to work process on FIFO(First In and First Out) basis.

Answer2:

Dispatcher receives the request from client and assigns the request to one of the work process.

Roll area: Each workprocess works in a particular memory that memory is known as Role Area, which consists of User context and session data.

ABAP- Processor :is an interpreter which can execute logic

Which one is not an exit comand ? (Exit, cencle, stop, back)

STOP.

Effect :The statement STOP is only to be used in executable programs

EXIT.

Effect :If the EXIT statement is executed outside of a loop, it will immediately terminate the current processing block.

BACK.

Effect : This statement positions the list cursor on the first position of the first line in a logical unit.

So "Cancle" is not an exit command

What is Field symbol ?

Answer1:

You can use field symbols to make the program more dynamic. In this example the name of a table control is substituted by a field symbol. Thus you can call the form with any internal table, using the name of the table control as a parameter.

Example

form insert_row

using p_tc_name.

field-symbols <tc> type cxtab_control. "Table control

assign (p_tc_name) to <tc>.

* insert 100 lines in table control

<tc>-lines = 100.

Answer2:

fieldsymbol has the same concept as pointer in c,

fieldsymbol don't point to a data type like char, num instead of that it points to the memory block. the syntax for fieldsymbol is

FIELD-SYMBOL <N>.

EG. FOR FIELD SYMBOL.

DATA: DAT LIKE SY-DATUM,

TIM LIKE SY-UZEIT,

CHAR(3) TYPE C VALUE 'ADF'.

FIELD-SYMBOL : <FS>.

MOVE DAT TO <FS>.

WRITE:/ <FS>.

MOVE TIM TO <FS>.

WRITE:/ <FS>.

MOVE CHAR TO <FS>.

WRITE:/ <FS>.

The output will be

Today's date

current time

What is lock object ?

LockObjects used to synchornize access of several users using same data.

Why BAPI need then BDC ?

BAPI'S provide the standard interface to other applications apart from SAP and within differnt vesions of SAP too. Also it is OOD bases so dosen't depends on screen flow. BDC gets failed if we make changes for screen changes through IMG customization

What are the advantages and disadvantages of using views in ABAP programming ?

advantages: view is used to retrieve the data very fastly from the database tables

- *memory wastage is reduced

- *faster than joins to retrieve the data from database tables

disadvantages:

view is not a container,it will not hold the data

- *view memory is not permanent memory

How data is stored in cluster table?

A cluster table conatins data from mulitple DDIC tables.

It stores data as a name value pair (varkey, vardata)

Have you used performance tuning? What major steps will you use for these?

First of all tunning can be done

In three ways: disk i/o ,sql tunning , memory tunning,

Before tunning u have to get the status of your database using

Oracle utility called statpack , tkprof, then you should go for tunning

How to create client independent tables?

client independent tables:

the table in which the first field is not mandt is the client independent tables

- *mandt is the field with mandt as the data element

- *automatically client which we login is populated to mandt

What type of user exits have you written?

there are four types

1. function exit

2. menu igit

3. screen exit.

4. field exit.

these are the user exits

What is a View ?

- A view is a logical grouping of one or more tables. A view on one or more tables i.e, the data from a view is not actually physically stored instead being derived from one or more tables. A view can be used to summarize data which is distributed among several tables

How are the types of Views?

Database View (SE11)

Database views are implement an inner join, that is, only records of the primary table (selected via the join operation) for which the corresponding records of the secondary tables also exist are fetched. Inconsistencies between primary and secondary table could, therefore, lead to a reduced selection set. In database views, the join conditions can be formulated using equality relationships between any base fields. In the other types of view, they must be taken from existing foreign keys. That is, tables can only be collected in a maintenance or help view if they are linked to one another via foreign keys.

Help View (SE54)

Help views are used to output additional information when the online help system is called. When the F4 button is pressed for a screen field, a check is first made on whether a matchcode is defined for this field. If this is not the case, the help view is displayed in which the check table of the field is the primary table. Thus, for each table no more than one help view can be created, that is, a table can only be primary table in at most one help view.

Projection View

Projection views are used to suppress or mask certain fields in a table (projection), thus minimizing the number of interfaces. This means that only the data that is actually required is exchanged when the database is accessed. A projection view can draw upon only one table. Selection conditions cannot be specified for projection views.

Maintenance View (SE54)

Maintenance views enable a business-oriented approach to looking at data, while at the same time, making it possible to maintain the data involved. Data from several tables can be summarized in a maintenance view and maintained collectively via this view. That is, the data is entered via the view and then distributed to the underlying tables by the system.

What is Locking ?

- When two users simultaneously attempt to access the same data record, this is synchronized by a lock mechanism.

When dialog transactions are programmed, locks are set and released by calling certain function modules. These function modules are generated automatically from the definition of so-called lock objects in the ABAP/4 Dictionary. To synchronize the access to a table by setting and removing locks, a Lock object has to be defined in the ABAP/4 Dictionary. Activating the lock object automatically creates #function modules for setting and removing locks. These function modules must be included when programming interactive transactions.

Lock Mechanism

To set locks, a lock object must be defined in the ABAP/4 Dictionary. In this lock object, those tables in which data records are to be locked by calling a lock are determined. All tables included in a lock object must be connected to each other via foreign keys. The key fields of the tables in a lock object form the Lock arguments for the tables. The lock arguments are the basis for formulating the logical condition for identifying the records to be locked. When activating this lock object, two function modules with the names ENQUEUE_ and DEQUEUE_ are generated.

Example :

Problem: You wish to prevent a user from being able to change the name of a course or the name of the professor with responsibility for the course at a time when another user is editing the course description (which contains this information).

Solution: The problem described above can be solved by defining a lock object E_UKURS. This is done by defining primary and secondary tables in the lock object. Table UKURS is check table of table UKRSB, so UKURS should be selected as primary table and UKRSB as secondary table of the lock object. The Lock argument in this case is the field combination FABNR, KRSNR, and SPRAS (i.e Primary Key Combination). The Lock mode Shared is to be selected here. This allows several users to access the data simultaneously in display mode. The lock mode in the generated function modules for setting (ENQUEUE_E_UKURS) and releasing (DEQUEUE_E_UKURS) locks is therefore set to shared as default, but can be overridden by calling the function modules. If the function module ENQUEUE_E_UKURS is called with FABNR = '1' and KRSNR = '3', the record for course 3 in faculty 1 is locked in table UKURS. Furthermore, all the course descriptions for this course are locked in table UKRSB since field SPRAS was not specified when the function module was called. In such cases, the lock is made generically for a field which is not defined. If the function module DEQUEUE_E_UKURS is now called with FABNR = '1', KRSNR = '3' and SPRAS = 'D', the German course description is unlocked. All other course descriptions remain locked.

What is database utility ?

Database utility is the interface between the ABAP/4 Dictionary and the underlying the SAP system.

The database utility is the interface between the ABAP/4 Dictionary and the relational database underlying the SAP system. You can call the database utility from the initial screen of the ABAP/4 Dictionary with Utilities ® Database utility. The database utility allows you to create, delete and convert objects from the ABAP/4 Dictionary in the database. MODULARIZATION What is Modularization and its benefits? If the program contains the same or similar blocks of statements or it is required to process the same function several times, we can avoid redundancy by using modularization techniques. By modularizing the ABAP/4 programs we make them easy to read and improve their structure. Modularized programs are also easier to maintain and to update.

1. What guarantees the integration of all application modules?

The R/3 basis system guarantees the integration of all application modules. The R/3 basis s/w provides the run time environment for the R/3 applications ensures optimal integration, defines a stable architectural frame for system enhancements, and contains the administration tools for the entire system. One of the main tasks of the basis system is to guarantee the portability of the complete system.

2. What are the central interfaces of the R/3 system?

Presentation Interface.
Database Interface.
Operating system Interface.

3. Which interface controls what is shown on the p.c.?

Presentation Interface.

4. Which interface converts SQL requirements in the SAP development system to those of the database?

Database Interface.

5. What is SAP dispatcher?

SAP dispatcher is the control agent that manages the resources for the R/3 applications.

6. What are the functions of dispatcher?

Equal distribution of transaction load to the work processes.
Management of buffer areas in main memory.
Integration of the presentation levels.
Organization of communication activities.

7. What is a work process?

A work process is where individual dialog steps are actually processed and the work is done. Each work process handles one type of request.

8. Name various work processes of R/3 system?

Dialog or Online (processes only one request at a time).
Background (Started at a specific time)
Update (primary or secondary)
Enque (Lock mechanism).
Spool (generated online or during back ground processing for printing).

9. Explain about the two services that are used to deal with communication.

Message Service: Used by the application servers to exchange short internal messages, all system communications.

Gateway Service: Enables communication between R/3 and external applications using CPI-C protocol.

10. Which work process triggers database changes?

Update work process.

11. Define service (within R/3)?

A service is a process or group of processes that perform a specific system function and often provide an application-programming interface for other processes to call.

12. What are the roll and page areas?

Roll and page areas are SAP R/3 buffers used to store user contexts (process requests). The SAP dispatcher assigns process requests to work processes as they are queued in the roll and page areas.

Paging area holds data from the application programs.

Roll area holds data from previous dialog steps and data that characterize the user.

13. What are the different layers in R/3 system?

Presentation Layer.

Application Layer.

Database Layer.

14. What are the phases of background processing?

Job Scheduling.

Job Processing.

Job Overview.

15. What components of the R/e system initiate the start of background jobs at the specified time?

The batch scheduler initiates the start of background job. The dispatcher then sends this request to an available background work process for processing.

16. Define Instance.

An instance is an administrative unit in which components of an R/3 systems providing one or more services are grouped together. The services offered by an instance are started and stopped at random. All components are parameterized using a joint instance profile. A central R/3 system consists of a single instance in which all-necessary SAP services are offered. Each instance uses separate buffer areas.

17. From hardware perspective, every information system can be divided into three task areas Presentation, Application Logic and Data Storage.

The R/3 Basis software is highly suitable for use in multi-level client/server architectures.

18. What are R/3 Basis configurations?

A central system with centrally installed presentation software.

Two-level client/server system with rolled out presentation software.

Two-level client/server system. Presentation and Application run on the same computer.

Three-level client/server system. Presentation, Application and database each run on separate computers.

19. What is a Service in SAP terminology?

A service refers to something offered by a s/w component.

20. What is Server in SAP terminology?

A component can consist of one process or a group and is then called the server for the respective service.

21. What is a client in SAP terminology?

A S/W component that uses the service (offered by a s/w component) is called a Client. At the same time these clients may also be servers for other services.

22. What is a SAP system?

The union of all s/w components that are assigned to the same databases is called as a SAP system.

23. What is the means of communications between R/3 and external applications?

The means of communication between R/2,R/3 and external applications is via the CPI-C handler or SAP Gateway, using the CPI-C Protocol.

24. What is the protocol used by SAP Gateway process?

The SAP Gateway process communicates with the clients based on the TCP/IP Protocol.

25. Expand CPI-C.

Common Program Interface Communication.

26. What is a Spool request?

Spool requests are generated during dialog or background processing and placed in the spool database with information about the printer and print format. The actual data is placed in the Tem Se (Temporary Sequential objects).

27. What are different types of Log records?

V1 and V2. V1 must be processed before V2. But, we can have more than one V2 logs.

28. What are the types of Update requests?

An update request can be divided into one primary (V1) and several Secondary update components (V2). Time-critical operations are placed in V1 component and those whose timing is less critical are placed in V2 components. If a V1 update fails, V2 components will not be processed.

29. Dialog work processes perform only one dialog step and then available for the next request.

30. Explain what is a transaction in SAP terminology.

In SAP terminology, a transaction is series of logically connected dialog steps.

31. Explain how SAP GUI handles output screen for the user.

The SAP front-end s/w can either run on the same computer or on different computers provided for that purpose. User terminal input is accepted by the SAP terminal program SAP GUI, converted to SAP proprietary format

and sent to the SAP dispatcher. The dispatcher coordinates the information exchange between the SAP GUIs and the work processes. The dispatcher first places the processing request in request queues, which it then processes. The dispatcher dispatches the requests one after another, to the available work process. The actual processing takes place in the work process. When processing is complete, the result of a work process is returned via the dispatcher to the SAP GUI. The SAP GUI interprets the received data and generates the output screen for the user.

BDC

1. What is full form of BDC Session?

Batch Data Communication Session.

2. What are the steps in a BDC session?

The first step in a BDC session is to identify the screens of the transaction that the program will process. Next step is to write a program to build the BDC table that will be used to submit the data to SAP. The final step is to submit the BDC table to the system in the batch mode or as a single transaction by the CALL TRANSACTION command.

3. How do you find the information on the current screen?

The information on the current screen can be found by SYSTEM à STATUS command from any menu.

4. How do you save data in BDC tables?

The data in BDC tables is saved by using the field name 'BDC_OKCODE' and field value of '/11'.

5. What is the last entry in all BDC tables?

In all BDC tables the last entry is to save the data by using the field name BDC_OKCODE and a field value of '/11'.

6. What is a multiple line field?

A multiple line field is a special kind of field which allows the user to enter multiple lines of data into it.

7. How do you populate data into a multiple line field?

To populate data into a multiple line field, an index is added to the field name to indicate which line is to be populated by the BDC session (Line index).

8. Write the BDC table structure.

BDC table structure

FIELD	TYPE	DESCRIPTION
Program	CHAR (8)	Program name of transaction.
DynPro	CHAR (4)	Screen number of transaction.
DynBegin	CHAR (1)	Indicator for new screen.
Fnam	CHAR (35)	Name of database field from screen.
Fval	CHAR (80)	Value to submit to field.

9. Does the CALL TRANSACTION method allow multiple transactions to be processed by SAP?

No. The CALL TRANSACTION method allows only a single transaction to be processed by SAP.

10. Does the BDC-INSERT function allow multiple transactions to be processed by SAP?

Yes.

11. What is the syntax for 'CALL TRANSACTION'?

CALL TRANSACTION trans [using bdctab MODE mode].

Three possible entries are there for MODE.

A	-	Show all screens.
E	-	Show only screens with errors.
N	-	Show no screens.

What are the layers of data description in R/3?

- ☐ The external layer.
- ☐ The ABAP/4 layer.
- ☐ The database layer.

Define external layer?

The external layer is the plane at which the user sees and interacts with the data, that is, the data format in the user interface. This data format is independent of the database system used.

Define ABAP/4 layer?

The ABAP/4 layer describes the data formats used by the ABAP/4 processor.

Define Database layer?

The database layer describes the data formats used in the database.

What is a Data Class?

The Data class determines in which table space the table is stored when it is created in the database.

What is a Size Category?

The Size category describes the probable space requirement of the table in the database.

How many types of size categories and data classes are there?

There are five size categories (0-4) and 11 data classes only three of which are appropriate for application tables:

- ☐ APPL0- Master data (data frequently accessed but rarely updated).
- ☐ APPL1- Transaction data (data that is changed frequently).
- ☐ APPL2- Organizational data (customizing data that is entered when system is configured and then rarely changed).

The other two types are:

- ☐ USR
- ☐ USR1 – Intended for customer's own developments.

What are control tables?

The values specified for the size category and data class are mapped to database-specific values via control tables.

What is the function of the transport system and workbench organizer?

The function of the transport system and the Workbench Organizer is to manage any changes made to objects of the ABAP/4 Development Workbench and to transport these changes between different SAP systems.

What is a table pool?

A table pool (or pool) is used to combine several logical tables in the ABAP/4 Dictionary. The definition of a pool consists of at least two key fields and a long argument field (VARDATA).

What are pooled tables?

These are logical tables, which must be assigned to a table pool when they are defined. Pooled tables can be used to store control data (such as screen sequences or program parameters).

What is a table cluster?

A table cluster combines several logical tables in the ABAP/4 Dictionary. Several logical rows from different cluster tables are brought together in a single physical record. The records from the cluster tables assigned to a cluster are thus stored in a single common table in the database.

How can we access the correction and transport system?

Each time you create a new object or change an existing object in the ABAP/4 Dictionary, you branch automatically to the Workbench Organizer or correction and transport system.

Which objects are independent transport objects?

Domains, Data elements, Tables, Technical settings for tables, Secondary indexes for transparent tables, Structures, Views, Matchcode objects, Matchcode Ids, Lock objects.

How is conversion of data types done between ABAP/4 & DB layer?

Conversion between ABAP/4 data types and the database layer is done within the database interface.

How is conversion of data types done between ABAP/4 & external level?

Conversion between the external layer and the ABAP/4 layer is done in the SAP dialog manager DYNP.

What are the Data types of the external layer?

ACCP, Char, CLNT, CUKY, CURR, DATS, DESC, FLTP, INT1, INT2, INT4, LANG, LCHR, LRAW, NUMC, PREC, QUAN, RAW, TMS, UNIT, VARC.

What are the Data types of the ABAP/4 layer?

Possible ABAP/4 data types:

C: Character.

D: Date, format YYYYMMDD.

F: Floating-point number in DOUBLE PRECISION (8 bytes).

I: Integer.

N: Numerical character string of arbitrary length.

P: Amount of counter field (packed; implementation depends on h/w platform).

S: Time Stamp YYYYMMDDHHMMSS.

V: Character string of variable length, length is given in the first two bytes.

X: Hexadecimal (binary) storage.

How can we set the table spaces and extent sizes?

You can specify the extent sizes and the table space (physical storage area in the database) in which a transparent table is to be stored by setting the size category and data class.

What is the function of the correction system?

The correction system manages changes to internal system components. Such as objects of the ABAP/4 Dictionary.

What are local objects?

Local objects (Dev class\$TMP) are independent of correction and transport system.

What is a Development class?

Related objects from the ABAP/4 repository are assigned to the same development class. This enables you to correct and transport related objects as a unit.

What is a data dictionary?

Data Dictionary is a central source of data in a data management system. Its main function is to support the creation and management of data definitions. It has details about

- ☐ what data is contained?
- ☐ What are the attributes of the data?
- ☐ What is the relationship existing between the various data elements?

What functions does a data dictionary perform?

In a data management system, the principal functions performed by the data dictionary are

- ☐ Management of data definitions.
- ☐ Provision of information for evaluation.
- ☐ Support for s/w development.
- ☐ Support form documentation.
- ☐ Ensuring that the data definitions are flexible and up-to-date.

What are the features of ABAP/4 Dictionary?

The most important features are:

- ☐ Integrated to aABAP/4 Development Workbench.
- ☐ Active in the runtime environment.

What are the uses of the information in the Data dictionary?

The following information is directly taken from the Data dictionary:

- ☐ Information on fields displayed with F1 help.
- ☐ Possible entries for fields displayed with F4 help.
- ☐ Matchcode and help views search utilities.

What are the basic objects of the data dictionary?

- ☐ Tables
- ☐ Domains
- ☐ Data elements
- ☐ Structures
- ☐ Foreign Keys

What are the aggregate objects in the data dictionary?

- ☐ Views
- ☐ Match codes
- ☐ Lock objects.

In the ABAP/4 Dictionary Tables can be defined independent of the underlying database (T/F).

True.

ABAP/4 Dictionary contains the Logical definition of the table.

A field containing currency amounts (data type CURR) must be assigned to a reference table and a reference field. Explain.

As a reference table, a system containing all the valid currencies is assigned or any other table, which contains a field with the currency key format. This field is called as reference field. The assignment of the field containing currency amounts to the reference field is made at runtime. The value in the reference field determines the currency of the amount.

A field containing quantity amounts (data type QUAN) must be assigned to a reference table and a reference field. Explain?

As a reference table, a system table containing all the valid quantity units is assigned or any other table, which contains a field with the format or quantity units (data type UNIT). This field is called as reference field.

The assignment of the field containing quantity amounts to the reference field is made at runtime. The value in the reference field determines the quantity unit of the amount.

What is the significance of Technical settings (specified while creating a table in the data dictionary)?

By specifying technical settings we can control how database tables are created in the database. The technical settings allows us to

- ☐ Optimize storage space requirements.
- ☐ Table access behavior.
- ☐ Buffering required.
- ☐ Changes to entries logged.

What is a Table attribute?

The table's attributes determine who is responsible for maintaining a table and which types of access are allowed for the table. The most important table attributes are:

- ☐ Delivery class.
- ☐ Table maintenance allowed.
- ☐ Activation type.

What is the significance of Delivery Class?

- ☐ The delivery class controls the degree to which the SAP or the customer is responsible for table maintenance.
- ☐ Whether SAP provides the table with or without contents.
- ☐ Determines the table type.
- ☐ Determines how the table behaves when it is first installed, at upgrade, when it is transported, and when a client copy is performed.

What is the max. no. Of structures that can be included in a table or structure.

Nine.

What are two methods of modifying SAP standard tables?

- ☐ Append Structures and
- ☐ Customizing Includes.

What is the difference between a Substructure and an Append Structure?

- ☐ In case of a substructure, the reference originates in the table itself, in the form of a statement include....
- ☐ In case of an append structure, the table itself remains unchanged and the reference originates in the append structure.

To how many tables can an append structure be assigned.

One.

If a table that is to be extended contains a long field, we cannot use append structures why?

Long fields in a table must always be located in the end, as the last field of the table. If a table has an append structure the append line must also be on the last field of the table.

Can we include customizing include or an append structure with Pooled or Cluster tables?

No.

What are the two ways for restricting the value range for a domain?

- ☐ By specifying fixed values.
- ☐ By stipulating a value table.

Structures can contain data only during the runtime of a program (T/F)

True.

What are the aggregate objects in the Dictionary?

- ☐ Views
- ☐ Match Code.
- ☐ Lock Object.

What are base tables of an aggregate object?

The tables making up an aggregate object (primary and secondary) are called aggregate object.

The data of a view is not physically stored, but derived from one or more tables (t/f)

True.

What are the 2 other types of Views, which are not allowed in Release 3.0?

- ☐ Structure Views.
- ☐ Entity Views.

What is a Match Code?

Match code is a tool to help us to search for data records in the system. Match Codes are an efficient and user-friendly search aid where key of a record is unknown.

What are the two levels in defining a Match Code?

- ☐ Match Code Object.
- ☐ Match Code Id.

What is the max no of match code Id's that can be defined for one Match code object?

A match code Id is a one character ID that can be a letter or a number.

Can we define our own Match Code ID's for SAP Matchcodes?

Yes, the number 0 to 9 are reserved for us to create our own Match Code Ids for a SAP defined Matchcode object.

What is an Update type with reference to a Match code ID?

If the data in one of the base tables of a matchcode ID changes, the matchcode data has to be updated. The update type stipulates when the matchcode is to be updated and how it is to be done. The update type also specifies which method is to be used for Building matchcodes. You must specify the update type when you define a matchcode ID.

Can matchcode object contain Ids with different update types?

Yes.

What are the update types possible?

The following update types are possible:

- ☐ Update type A: The matchcode data is updated asynchronously to database changes.
- ☐ Update type S: The matchcode data is updated synchronously to database changes.
- ☐ Update type P: The matchcode data is updated by the application program.
- ☐ Update type I: Access to the matchcode data is managed using a database view.
- ☐ Update type L: Access to the matchcode is achieved by calling a function module.

What are the two different ways of building a match code object?

A match code can be built in two different ways:

- ☐ Logical structure: The matchcode data is set up temporarily at the moment when the match code is accessed. (Update type I, k).
- ☐ Physical Structure: The match code data is physically stored in a separate table in the database. (Update type A, S, P).

What are the differences between a Database index and a match code?

- ☐ Match code can contain fields from several tables whereas an index can contain fields from only one table.
- ☐ Match code objects can be built on transparent tables and pooled and cluster tables.

What is the function of a Domain?

- ☐ A domain describes the technical settings of a table field.
- ☐ A domain defines a value range, which sets the permissible data values for the fields, which refers to this domain.
- ☐ A single domain can be used as basis for any number of fields that are identical in structure.

Can you delete a domain, which is being used by data elements?

No.

What are conversion routines?

- ☐ Non-standard conversions from display format to sap internal format and vice-versa are implemented with so called conversion routines.

What is the function of a data element?

A data element describes the role played by a domain in a technical context. A data element contains semantic information.

Can a domain, assigned to a data element be changed?

Yes. We can do so by just overwriting the entry in the field domain.

Can you delete data element, which is being used by table fields.

No.

Can you define a field without a data element?

Yes. If you want to specify no data element and therefore no domain for a field, you can enter data type and field length and a short text directly in the table maintenance.

What are null values?

If the value of a field in a table is undefined or unknown, it is called a null value.

What is the difference between a structure and a table?

Structures are constructed the almost the same way as tables, the only difference using that no database table is generated from them.

What is a view?

A view is a logical view on one or more tables. A view on one or more tables i.e., the data from a view is not actually physically stored instead being derived from one or more tables.

How many types of Views are there?

- ☐ Database View
- ☐ Help View
- ☐ Projection View
- ☐ Maintenance View

What is Locking?

When two users simultaneously attempt to access the same data record, this is synchronized by a lock mechanism.

What is database utility?

Database utility is the interface between the ABAP/4 Dictionary and the underlying the SAP system.

What are the basic functions of Database utility?

The basic functions of database utility are:

- ☐ Create database objects.
- ☐ Delete database objects.
- ☐ Adjust database objects to changed ABAP/4 dictionary definition.

What is Repository Info. Systems?

It is a tool with which you can make data stored in the ABAP/4 Dictionary available.

Does every ABAP/4 have a modular structure?

Yes.

What is Modularization and its benefits?

If the program contains the same or similar blocks of statements or it is required to process the same function several times, we can avoid redundancy by using modularization techniques. By modularizing the ABAP/4 programs we make them easy to read and improve their structure. Modularized programs are also easier to maintain and to update.

Name the ABAP/4 Modularization techniques.

- Source code module.
- Subroutines.
- Functions.

How can we create callable modules of program code within one ABAP/4 program?

- By defining Macros.
- By creating include programs in the library.

M is the attribute type of the module program.

Is it possible to pass data to and from include programs explicitly?

No. If it is required to pass data to and from modules it is required to use subroutines or function modules.

What are subroutines?

Subroutines are program modules, which can be called from other ABAP/4 programs or within the same program.

What are the types of Subroutines?

- Internal Subroutines: The source code of the internal subroutines will be in the same ABAP/4 program as the calling procedure (internal call).
- External Subroutines: The source code of the external subroutines will be in an ABAP/4 program other than the calling procedure.

It is not possible to create an ABAP/4 program, which contains only Subroutines (T/F).

False.

A subroutine can contain nested form and endform blocks. (T/F)

False.

Data can be passed between calling programs and the subroutines using Parameters.

What are the different types of parameters?

Formal Parameters: Parameters, which are defined during the definition of subroutine with the FORM statement.

Actual Parameters: Parameters which are specified during the call of a subroutine with the PERFORM statement.

How can one distinguish between different kinds of parameters?

- Input parameters are used to pass data to subroutines.
- Output parameters are used to pass data from subroutines.

What are the different methods of passing data?

- Calling by reference: During a subroutine call, only the address of the actual parameter is transferred to the formal parameters. The formal parameter has no memory of its own, and we work with the field of the calling program within the subroutine. If we change the formal parameter, the field contents in the calling program also changes.

- Calling by value: During a subroutine call, the formal parameters are created as copies of the actual parameters. The formal parameters have memory of their own. Changes to the formal parameters have no effect on the actual parameters.

- Calling by value and result: During a subroutine call, the formal parameters are created as copies of the actual parameters. The formal parameters have their own memory space. Changes to the formal parameters are copied to the actual parameters at the end of the subroutine.

The method by which internal tables are passed is By Reference.

How can an internal table with Header line and one without header line be distinguished when passed to a subroutine?

Itab [] is used in the form and endform if the internal table is passed with a header line.

What should be declared explicitly in the corresponding ABAP/4 Statements to access internal tables without header lines & why?

Work Area. This is required as the Work Area is the interface for transferring data to and from the table.

A subroutine can be terminated unconditionally using EXIT. (T/F)

True.

A subroutine can be terminated upon a condition using CHECK Statement.

Function Modules are also external Subroutines. (T/F).

True.

What is the difference between the function module and a normal ABAP/4 subroutine?

In contrast to normal subroutines function modules have uniquely defined interface. Declaring data as common parts is not possible for function modules. Function modules are stored in a central library.

What is a function group?

A function group is a collection of logically related modules that share global data with each other. All the modules in the group are included in the same main program. When an ABAP/4 program contains a CALL FUNCTION statement, the system loads the entire function group in with the program code at runtime. Every function module belongs to a function group.

What is the disadvantage of a call by reference?

During a call by reference damage or loss of data is not restricted to the subroutine, but will instantly lead to changes to the original data objects.

A function module can be called from a transaction screen outside an ABAP/4 program. (T/F).

True.

What is an update task?

It is an SAP provided procedure for updating a database.

What happens if a function module runs in an update task?

The system performs the module processing asynchronously. Instead of carrying out the call immediately, the system waits until the next database update is triggered with the 'COMMIT WORK' command.

The function modules are created and stored in the Function Library.

When a function module is activated syntax checking is performed automatically. (Y/N)

True.

What is the use of the RAISING exception?

The raising exception determines whether the calling program will handle the exception itself or leave the exception to the system.

What is the difference between internal tables and extract datasets?

- The lines of an internal table always have the same structure. By using extract datasets, you can handle groups of data with different structure and get statistical figures from the grouped data.

- You have to define the structure of the internal table at the beginning. You need not define the structure of the extract dataset.

- In contrast to internal tables, the system partly compresses exact datasets when storing them. This reduces the storage space required.

- Internal tables require special work area for interface whereas extract datasets do not need a special work area for interface.

It is possible to assign a local data object defined in a subroutine or function module to a field group. (T/F).

False.

What is the difference between field-group header and other field groups?

The header field group is a special field group for the sort criteria. The system automatically prefixes any other field groups with the header field group.

Can a field occur in several field groups.

Yes. But it leads to unnecessary data redundancy.

What does the extract statement do in extract datasets?

The data is written to virtual memory by extract commands.

A field-groups statement or an insert statement reverses storage space and transfers values. (T/F).

False.

While using extract datasets it is required to have a special workarea for interface (T/F)

False.

The LOOP-ENDLOOP on extract datasets can be used without any kind of errors (T/F)

False. It causes runtime errors.

The Maximum no of key fields that can be used in a header is 50.

While sorting field groups we cannot use more than one key field (T/F).

False.

While sorting, if the main storage available is not enough, the system writes data to an external help file. The SAP profile parameter, which determines this help file, is DIR_SORTTMP.

Write:s_time using edit mask'--: ---- '.

In order to suppress the leading zeroes of a number field the keywords used are NO-ZERO.

The total no of date formats that can be used to display a date during output is MM/DD/YY, DD/MM/YY, DD/MM/YYYY, MM/DD/YYYY, MMDDYY, DDMMYY, YYMMDD.

The UNDER Command allows for vertical alignment of fields one below the other.

In order to concatenate strings only for output purposes the command NO-GAP can be used in conjunction with the 'Write' statement.

How would you format the output as left, centered or right-justified using the write statement.

Left-justified, Centered, Right-justified.

If the same formatting options were used for a WRITE statement that follows the FORMAT statement, which settings would take precedence.

The settings in the Write Statement.

For each new event, the system resets all formatting options to their default values (T/F)

TRUE.

To execute a page break under the condition that less than a certain number of lines is left on a page is achieved by **RESERVE** n lines.

The RESERVE statement only takes effect if output is written to the subsequent page. No blank pages are created and it defines a block of lines that must be output as a whole. (T/F). TRUE.

To set the next output line to the first line of a block of lines defined with the RESERVE statement the statement **BACK** is used.

What is the limit for the length of a page if the page length is not specified in the report statement. 60,000 lines.

How would you start the printing process from within the program while creating a list?

NEW-PAGE PRINT ON.

You can change the width of pages within list levels triggered by page breaks. (T/F).

FALSE.

Hotspots are special areas of an output list used to trigger events. (T/F) TRUE.

To designate fields as hotspots at runtime, use **FORMAT HOTSPOT = <h>**.

How would you suppress the display of a parameter on the selection screen?

Parameters <p>.....No-Display.

Can you assign a matchcode object to a parameter? If so how?

Yes. **PARAMETERS <p>.....MATCHCODE OBJECT <obj>.....**

For each SELECT-OPTIONS statement, the system creates a selection table. (T/F)

TRUE.

To position a set of parameters or comments on a single line on the selection screen, you must declare the elements in a block enclosed by

SELECTION-SCREEN BEGIN OF LINE.

.....

SELECTION-SCREEN END OF LINE.

How can Symbols or R/3 icons be output on the screen?

WRITE <symbol-name>AS SYMBOL.

WRITE <icon-name> AS ICON.

In the standard setting, you cannot create empty lines with the WRITE statement alone. (T/F). TRUE.

REPORTING – GENERAL

The system field, which indicates success or failure of a SQL operation, is SY-SUBRC.

What is the syntax for specifying database table name at runtime in SELECT statement.

NAME = 'SPFL1'.

SELECT * FROM (NAME).

.....

ENDSELECT.

How do you read selected lines of database table into an internal table in packages of predefined size.

SELECT * FROM <SPFLI>INTO TABLE <ITAB>PACKAGE SIZE<N>.....Where n is variable.

In SELECT statements can you specify a variable in WHERE condition or a part of the condition, if so what is the syntax.

SELECT * FROM <table>WHERE <var1><condition><var or const>.

Name the ABAP/4 key words, which are used to change the contents of database table.

UPDATE or MODIFY.

How to specify a client for database table processing.

TABLES SPFLI.

SELECT * FROM SPFLI CLIENT SPECIFIED WHERE MANDT BETWEEN '001' AND '003'.

.....

ENDSELECT.

How do you write a DATA object from ABAP/4 program to ABAP/4 memory and restore the same from memory to program.

EXPORT <f1>[FROM <g1>]<f2>[FROM <g2>]... TO MEMORY ID <key>.

The ID <key>, which can be up to 32 characters long, identifies the data in memory.

What are DATA CLUSTERS?

You can group any complex internal data objects of an ABAP/4 program together in data clusters and store them temporarily in ABAP/4 memory or for longer periods in databases. You can store data clusters in special databases of the ABAP/4 Dictionary. These databases are known as ABAP/4 cluster databases and have a predefined structure. Storing a data cluster is specific to ABAP/4. Although you can also access cluster databases using SQL statements, only ABAP/4 statements are able to decode the structure of the stored data cluster.

Statements used to delete data objects in ABAP/4 memory FREE MEMORY [ID <key>].

Name the ABAP/4 key word, which is used to clear the Headerline of an Internal Table.

CLEAR<itab>.

Name the function modules to read data from Presentation Server into an Internal Table.

UPLOAD and WS_UPLOAD.

Name the ABAP/4 keywords to initialize an Internal Table with and without headerline.

REFRESH <itab>.

How to determine the attributes of an internal table?

DESCRIBE TABLE <itab>[LINES <lin>] [OCCURS <occ>].

Name the ABAP/4 key word for searching a string in an Internal Table.

SEARCH <itab> FOR <str><options>.

The different options (<options>) for the search in an internal table are:

ABBREVIATED

Searches table<itab>for a word containing the character string specified in <str>, where other characters might separate the characters. The first letter of the word and the string <str> must be the same.

STARTING AT<lin1>

Searches table<itab> for <str>, starting at line <line1>. <lin1> can be a variable.

ENDING AT<n2>

Searches table <itab>for <str>upto line<lin2>. <lin2>can be a variable.

AND MARK

If the search string is found, all the characters in the search string (and all the characters in between when using ABBREVIATED) are converted to upper case.

What are the different attributes that can be assigned to a variant?

The different attributes that can be assigned to a variant are....

Description

Enter a short, meaningful description of the variant. This may be upto 30 characters long.

Background only

Specify whether you want to use the variant in background processing only, or in online environment as well.

Protected variant.

Mark the field if you want to protect your variant against being changed by other users.

Do not display variant.

Mark this field if you want the variant name to be displayed in the catalog only, but not in the F4 value list.

For the selections you cover in a variant, you can enter the following attributes:

Type

The system displays whether the field is a parameter or a select option.

Protected

Mark this field for each field on the selection screen you want to protect from being overwritten. Values that you mark this way are displayed to the users, but they cannot change them, that are they are not ready to accept input.

Invisible

If you mark this column, the system will not display the corresponding field on the selection screen the user sees when starting the report program.

Variable

Mark this column if you want to set the value for this field at runtime.

Is it possible to create new dynamic programs during runtime of an ABAP/4 program? If so how?

To create new dynamic programs during the runtime of an ABAP/4 program, you must use an internal table. For this purpose, you should create this internal table with one character type column and a line width of 72. You can use any method you like from Filling Internal Tables to write the code of your new program into the internal table. Especially, you can use internal fields in which contents are dependent on the flow of the program that you use to create a new one, to influence the coding of the new program dynamically. The following example shows how to proceed in principal:

```
DATA CODE (72) OCCURS 10.
```

```
APPEND 'REPORT ZDYN1.'
```

```
TO CODE.
```

```
APPEND 'WRITE/"Hello, I am dynamically created!".'
```

```
TO CODE.
```

Two lines of a very simple program are written into the internal table CODE.

In the next step you have to put the new module, in the above example it is a report, into the library. For this purpose you can use the following statement:

Syntax

```
INSERT REPORT <prog>FROM <itab>.
```

The program <prog> is inserted in your present development class in the R/3 Repository. If a program with this name does not already exist, it is newly created with the following attributes:

Title: none,

Type: 1 (Reporting),

Application: S (Basis).

You can specify the name of the program <prog> explicitly within single quotation marks or you can write the name of a character field, which contains the program name. The name of the program must not necessarily be the same as given in the coding, but it is recommended to do so. <itab> is the internal table containing the source code. For the above example you could write:

```
INSERT REPORT 'ZDYN1' FROM CODE.
```

Or

```
DATA REP (8).
```

```
REP = 'ZDYN1'
```

```
INSERT REPORT REP FROM CODE.
```

The data object does not occupy any space in memory. (T/F)

FALSE.

What are the three hierarchical levels of data types and objects?

Program-independent data, defined in the ABAP/4 Dictionary.

Internal data used globally in one program.

Data used locally in a procedure (subroutine, function module)

Which function module would you use to check the user's authorization to access files before opening a file?

```
AUTHORITY_CHECK_DATASET
```

Name the function module used to convert logical file names to physical file names in ABAP/4 programs.

```
FILE_GET_NAME.
```

Parameters, which are defined during the definition of a subroutine with the FORM statement, are called **Formal Parameters**.

Parameters which are specified during the call of a subroutine with the PERFORM statement are called **Actual Parameters**.

In subroutines internal tables that are passed by TABLES, are always called by value and result. (T/F)

FALSE. They are called by reference.

INTERACTIVE REPORTING

What is interactive reporting?

It helps you to create easy-to-read lists. You can display an overview list first that contains general information and provide the user with the possibility of choosing detailed information that you display on further lists.

What are the uses of interactive reporting?

The user can actively control data retrieval and display during the session. Instead of an extensive and detailed list, you create a basic list with condensed information from which the user can switch to detailed displays by positioning the cursor and entering commands. The detailed information appears in secondary lists.

What are the event key words in interactive reporting?

Event Keyword	Event
AT LINE-SELECTION	Moment at which the user selects a line by double clicking on it or by positioning the cursor on it and pressing F2.
AT USER-COMMAND	Moment at which the user presses a function key.
TOP-OF-PAGE DURING LINE-SELECTION	Moment during list processing of a secondary list at which a new page starts.

What is secondary list?

It allows you to enhance the information presented in the basic list. The user can, for example, select a line of the basic list for which he wants to see more detailed information. You display these details on a secondary list. Secondary lists may either overlay the basic list completely or you can display them in an extra window on the screen. The secondary lists can themselves be interactive again.

How to select valid lines for secondary list?

To prevent the user from selecting invalid lines, ABAP/4 offers several possibilities. At the end of the processing block END-OF-SELECTION, delete the contents of one or more fields you previously stored for valid lines using the HIDE statement. At the event AT LINE-SELECTION, check whether the work area is initial or whether the HIDE statement stored field contents there. After processing the secondary list, clear the work area again. This prevents the user from trying to create further secondary lists from the secondary list displayed.

How to create user interfaces for lists?

The R/3 system automatically, generates a graphical user interface (GUI) for your lists that offers the basic functions for list processing, such as saving or printing the list. If you want to include additional functionality, such as pushbuttons, you must define your own interface status. To create a new status, the Development Workbench offers the Menu Painter. With the Menu Painter, you can create menus and application toolbars. And you can assign Function Keys to certain functions. At the beginning of the statement block of AT END-OF-SELECTION, active the status of the basic list using the statement: SET PF-STATUS 'STATUS'.

What is interactive reporting?

A classical non-interactive report consists of one program that creates a single list. Instead of one extensive and detailed list, with interactive reporting you create basic list from which the user can call detailed information by positioning the cursor and entering commands. Interactive reporting thus reduces information retrieval to the data actually required.

Can we call reports and transactions from interactive reporting lists?

Yes. It also allows you to call transactions or other reports from lists. These programs then use values displayed in the list as input values. The user can, for example, call a transaction from within a list of change the database table whose data is displayed in the list.

What are system fields for secondary lists?

SY-LSIND	Index of the list created during the current event (basic list = 0)	
SY-LISTI	Index of the list level from which the event was triggered.	
SY-LILLI	Absolute number of the line from which the event was triggered.	
SY-LISEL	Contents of the line from which the event was triggered.	
SY-CUROW	Position of the line in the window from which the event was triggered (counting starts with 1)	
SY-CUCOL	Position of the column in the window from which the event was triggered	(counting starts with 2).
SY-CPAGE	Page number of the first displayed page of the list from which the event was triggered.	

SY-STARO Number of the first line of the first page displayed of the list from which the event was triggered (counting starts with 1). Possibly, a page header occupies this line.
SY-STACO Number of the first column displayed in the list from which the event was triggered (counting starts with 1).
SY-UCOMM Function code that triggered the event.
SY-PFKEY Status of the displayed list.

How to maintain lists?

To return from a high list level to the next-lower level (SY-LSIND), the user chooses Back on a secondary list. The system then releases the currently displayed list and activates the list created one step earlier. The system deletes the contents of the released list. To explicitly specify the list level, into which you want to place output, set the SY-LSIND field. The system accepts only index values, which correspond to existing list levels. It then deletes all existing list levels whose index is greater or equal to the index specify. For example, if you set SY-LSIND to 0, the system deletes all secondary lists and overwrites the basic list with the current secondary list.

What are the page headers for secondary lists?

On secondary lists, the system does not display a standard page header and it does not trigger the event. TOP-OF-PAGE. To create page headers for secondary list, you must enhance TOP-OF-PAGE: Syntax TOP-OF-PAGE DURING LINE-SELECTION. The system triggers this event for each secondary list. If you want to create different page headers for different list levels, you must program the processing block of this event accordingly, for example by using system fields such as SY-LSIND or SY-PFKEY in control statements (IF, CASE).

How to use messages in lists?

ABAP/4 allows you to react to incorrect or doubtful user input by displaying messages that influence the program flow depending on how serious the error was. Handling messages is mainly a topic of dialog programming. You store and maintain messages in Table T100. Messages are sorted by language, by a two-character ID, and by a three-digit number. You can assign different message types to each message you output. The influence of a message on the program flow depends on the message type. In our program, use the MESSAGE statement to output messages statically or dynamically and to determine the message type.

Syntax:REPORT <rep> MESSAGE-ID <id>.

What are the types of messages?

A message can have five different types. These message types have the following effects during list processing:

.A (=Abend):

.E (=Error) or W (=Warning):

.I (=Information):

.S (=Success):

What are the user interfaces of interactive lists?

If you want the user to communicate with the system during list display, the list must be interactive. You can define specific interactive possibilities in the status of the list's user interface (GUI). To define the statuses of interfaces in the R/3 system, use the Menu Painter tool. In the Menu Painter, assign function codes to certain interactive functions. After an user action occurs on the completed interface, the ABAP/4 processor checks the function code and, if valid, triggers the corresponding event.

What are the drill-down features provided by ABAP/4 in interactive lists?

ABAP/4 provides some interactive events on lists such as AT LINE-SELECTION (double click) or AT USER-COMMAND (pressing a button). You can use these events to move through layers of information about individual items in a list.

What is meant by stacked list?

A stacked list is nothing but secondary list and is displayed on a full-size screen unless you have specified its coordinates using the window command.

Is the basic list deleted when the new list is created?

No. It is not deleted and you can return back to it using one of the standard navigation functions like clicking on the back button or the cancel button.

What is meant by hotspots?

A Hotspot is a list area where the mouse pointer appears as an upright hand symbol. When a user points to that area (and the hand cursor is active), a single click does the same thing as a double-click. Hotspots are supported from R/3 release 3.0c.

What is the length of function code at user-command?

Each menu function, push button, or function key has an associated function code of length FOUR (for example, FREE), which is available in the system field SYUCOMM after the user action.

Can we create a gui status in a program from the object browser?

Yes. You can create a GUI STATUS in a program using SET PF-STATUS.

In which system field does the name of current gui status is there?

The name of the current GUI STATUS is available in the system field SY-PFKEY.

Can we display a list in a pop-up screen other than full-size stacked list?

Yes, we can display a list in a pop-up screen using the command WINDOW with the additions starting at X1 Y1 and ending at X2 Y2 to set the upper-left and the lower-right corners where x1 y1 and x2 y2 are the coordinates.

What is meant by hide area?

The hide command temporarily stores the contents of the field at the current line in a system-controlled memory called the HIDE AREA. At an interactive event, the contents of the field are restored from the HIDE AREA.

When the get cursor command used in interactive lists?

If the hidden information is not sufficient to uniquely identify the selected line, the command GET CURSOR is used. The GET CURSOR command returns the name of the field at the cursor position in a field specified after the addition field, and the value of the selected field in a field specified after value.

How can you display frames (horizontal and vertical lines) in lists?

You can display tabular lists with horizontal and vertical lines (FRAMES) using the ULINE command and the system field SY-VLINE. The corners arising at the intersection of horizontal and vertical lines are automatically drawn by the system.

What are the events used for page headers and footers?

The events TOP-OF-PAGE and END-OF-PAGE are used for pager headers and footers.

How can you access the function code from menu painter?

From within the program, you can use the SY-UCOMM system field to access the function code. You can define individual interfaces for your report and assign them in the report to any list level. If you do not specify self-defined interfaces in the report but use at least one of the three interactive event keywords. AT LINE-SELECTION, AT PF<nn>, OR AT USER-COMMAND in the program, the system automatically uses appropriate predefined standard interfaces. These standard interfaces provide the same functions as the standard list described under the standard list.

How the at-user command serves mainly in lists?

The AT USER-COMMAND event serves mainly to handle own function codes. In this case, you should create an individual interface with the Menu Painter and define such function codes.

How to pass data from list to report?

ABAP/4 provides three ways of passing data:

- Passing data automatically using system fields
- Using statements in the program to fetch data
- Passing list attributes

How can you manipulate the presentation and attributes of interactive lists?

- Scrolling through Interactive Lists.
- Setting the Cursor from within the Program.
- Modifying List Lines.

How to call other programs?

Report	Transaction
Call and return	SUBMIT AND RETURN CALL TRANSACTION
Call without return	SUBMIT LEAVE TO TRANSACTION

You can use these statements in any ABAP/4 program.

What will exactly the hide statement do?

For displaying the details on secondary lists requires that you have previously stored the contents of the selected line from within the program. To do this, ABAP/4 provides the HIDE statement. This statement stores the current field contents for the current list line. When calling a secondary list from a list line for which the HIDE fields are stored, the system fills the stored values back into the variables in the program. In the program code, insert the HIDE statement directly after the WRITE statement for the current line. Interactive lists provide the

user with the so-called 'INTERACTIVE REPORTING' facility. For background processing the only possible method of picking the relevant data is through 'NON INTERACTIVE REPORT' . After starting a background job, there is no way of influencing the program. But whereas for dialog sessions there are no such restrictions.

How many lists can a program produce?

Each program can produce up to 21 lists: one basic list and 20 secondary lists. If the user creates a list on the next level (that is, SY-LSIND increases), the system stores the previous list and displays the new one. Only one list is active, and that is always the most recently created list.

FALSE.

What is a transaction?

- A transaction is dialog program that change data objects in a consistent way.

What are the requirements a dialog program must fulfill?

A dialog program must fulfil the following requirements

- A user friendly user interface.
- Format and consistency checks for the data entered by the user.
- Easy correction of input errors.
- Access to data by storing it in the data bases.

What are the basic components of dialog program?

- Screens (Dynpros)
- Each dialog in an SAP system is controlled by dynpros. A dynpro consists of a screen and its flow logic and controls exactly one dialog step.

- ABAP/4 module Pool.

Each dynpro refers to exactly one ABAP/4 dialog program .Such a dialog program is also called a module pool ,since it consists of interactive modules.

What is PBO and PAI events?

PBO- Process Before Output-It determines the flow logic before displaying the screen.

PAI-Process After Input-It determines the flow logic after the display of the screen and after receiving inputs from the User.

What is dynpro?What are its components ?

- A dynpro (Dynamic Program) consists of a screen and its flow logic and controls exactly one dialog steps.
- The different components of the dynpro are :

Flow Logic: calls of the ABAP/4 modules for a screen .

Screen layout: Positions of the text, fields, pushbuttons and so on for a screen

Screen Attributes: Number of the screen, number of the subsequent screen, and others

Fields attributes: Definition of the attributes of the individual fields on a screen.

What is a ABAP/4 module pool?

- Each dynpro refers to exactly one ABAP/4 dialog program.Such a dialog program is also called a module pool ,since it consists on interactive modules.

Can we use WRITE statements in screen fields?if not how is data transferred from field data to screen fields?

-We cannot write field data to the screen using the WRITE statement.The system instead transfers data by comparing screen fields names with ABAP/4 variable names.If both names are the same,it transfers screen fields values to ABAP/4 programs fields and Vice Versa.This happens immediately after displaying the screen.

Can we use flow logic control key words in ABAP/4 and vice-versa?

- The flow control of a dynpro consists of a few statements that syntactically resemble ABAP/4 statements .However ,we cannot use flow control keywords in ABAP/4 and vice-versa.

What is GUI status? How to create /Edit GUI status?

-A GUI status is a subset of the interface elements used for a certain screen.The status comprises those elements that are currently needed by the transaction .The GUI status for a transaction may be composed of the following elements:

- Title bar.
- Menu bar.
- Application tool bar

-Push buttons.

To create and edit GUI status and GUI title, we use the Menu Painter.

How does the interaction between the Dynpro and the ABAP/4 Modules take place?

-A transaction is a collection of screens and ABAP/4 routines, controlled and executed by a Dialog processor. The Dialog processor processes screen after the screen, thereby triggering the appropriate ABAP/4 processing of each screen. For each screen, the system executes the flow logic that contains the corresponding ABAP/4 processing. The control passes from screen flow logic to ABAP/4 code and back.

How does the Dialog handle user requests?

- when an action is performed, the system triggers the PROCESS AFTER INPUT event. The data passed includes field screen data entered by the user and a function code. A function code is a technical name that has been allocated in a screen Painter or Menu Painter to a menu entry, a push button, the ENTER key or a function key of a screen. An internal work field (OK-CODE) in the PAI module evaluates the function code, and the appropriate action is taken.

What is to be defined for a push button fields in the screen attributes?

- A function code has to be defined in the screen attributes for the push buttons in a screen.

How are the function codes handled in Flow Logic?

- When the User selects a function in a transaction, the system copies the function code into a specially designated work field called OK_CODE. This field is Global in ABAP/4 Module Pool. The OK_CODE can then be evaluated in the corresponding PAI module. The function code is always passed in exactly the same way, regardless of whether it comes from a screen's pushbutton, a menu option, function key or other GUI element.

What controls the screen flow?

- The SET SCREEN and LEAVE SCREEN statements control screen flow.

The function code currently active is ascertained by what variable?

- The function code currently active in a Program can be ascertained from the SY-UCOMM Variable.

The function code currently active is ascertained by what variable?

- By SY-UCOMM Variable.

What are the "field" and "chain" statements?

- The FIELD and CHAIN flow logic statements let you Program Your own checks. FIELD and CHAIN tell the system which fields you are checking and whether the system should perform checks in the flow logic or call an ABAP/4 Module.

What is an "on input field" statement?

- ON INPUT

The ABAP/4 module is called only if a field contains the value other than the initial value. This initial value is determined by the field's data type: blanks for character fields, zeroes for numerics. If the user changes the field's value back to its initial value, ON INPUT does not trigger a call.

What is an "on request field" statement?

- ON REQUEST

The ABAP/4 Module is called only if the user has entered the value in the field value since the last screen display. The value counts as changed even if the user simply types in the value that was already there. In general, the ON REQUEST condition is triggered through any form of "MANUAL INPUT".

What is an "on *-input field" statement?

ON *-INPUT

- The ABAP/4 module is called if the user has entered the "*" in the first character of the field, and the field has the attribute *-entry in the screen Painter. You can use this option in exceptional cases where you want to check only fields with certain kinds of input.

What are conditional chain statements?

ON CHAIN-INPUT similar to ON INPUT.

The ABAP/4 module is called if any one of the fields in the chain contains a value other than its initial value (blank or nulls).

ON CHAIN-REQUEST

This condition functions just like ON REQUEST, but the ABAP/4 module is called if any one of the fields in the chain changes value.

What is “at exit-command”?

The flowlogic Keyword at EXIT-COMMAND is a special addition to the MODULE statement in the Flow Logic .AT EXIT-COMMAND lets you call a module before the system executes the automatic fields checks.

Which Function type has to be used for using “at exit-command” ?

- To Use AT EXIT – COMMAND ,We must assign a function Type “E” to the relevant function in the MENU Painter OR Screen Painter .

What are the different message types available in the ABAP/4 ?

- There are 5 types of message types available.
- E: ERROR
- W-WARNING
- I –INFORMATION
- A-ABNORMAL TERMINATION.
- S-SUCCESS

Of the two “ next screen “ attributes the attributes that has more priority is ----- --.
Dynamic.

Navigation to a subsequent screen can be specified statically/dynamically. (TRUE/FALSE).
TRUE.

Dynamic screen sequence for a screen can be set using ----- and -----commands
Set Screen, Call screen.

The commands through Which an ABAP/4 Module can “branch to “ or “call” the next screen are
Set screen<scr no>,Call screen<scr no> ,Leave screen, Leave to screen <scr no>.

What is difference between SET SCREEN and CALL SCREEN ?

- With SET SCREEN the current screen simply specifies the next screen in the chain , control branches to this next screen as soon as the current screen has been processed .Return from next screen to current screen is not automatic .It does not interrupt processing of the current screen.If we want to branch to the next screen without finishing the current one ,use LEAVE SCREEN.
- With CALL SCREEN , the current (calling) chain is suspended , and a next screen (screen chain) is called .The called can then return to the suspended chain with the statement LEAVE SCREEN TO SCREEN 0 .Sometime we might want to let an user call a pop up screen from the main application screen to let him enter secondary information.After they have completed their entries, the users should be able to close the popup and return directly to the place where they left off in the main screen.Here comes CALL SCREEN into picture .This statement lets us insert such a sequence into the current one.

Can we specify the next screen number with a variable (*Yes/No)?

- Yes
- The field SY-DYNR refers to: Number of the current screen.

What is dialog Module?

- A dialog Module is a callable sequence of screens that does not belong to a particular transaction.Dialog modules have their module pools , and can be called by any transaction.

The Syntax used to call a screen as dialog box (pop up)is: CALL SCREEN <screen number.>

STARTING AT <start column><start line>

ENDING AT <end column> <end line>

What is “call mode”?

- In the ABAP/4 WORLD each stackable sequence of screens is a “call mode”, This is IMP because of the way u return from the given sequence .To terminate a call mode and return to a suspended chain set the “next screen” to 0 and leave to it:

LEAVE TO SCREEN 0 or (SET SCREEN 0 and LEAVE SCREEN) .When u return to the suspended chain execution resumes with the statement directly following the original CALL SCREEN statement.The original sequence of screens in a transaction (that is , without having stacked any additional call modes),you returned from the transaction altogether.

The max number of calling modes stacked at one time is?

- NINE

What is LUW or Data base Transaction ?

- A “LUW”(logical unit of work) is the span of time during which any database updates must be performed in an “all or nothing” manner .Either they are all performed (committed),or they are all thrown away (rolled back).In the ABAP/4 world , LUWs and

- Transactions can have several meanings:

LUW (or “database LUW” or “database transaction”)

This is the set of updates terminated by a database commit. A LUW lasts, at most, from one screen change to the next (because the SAP system triggers database commits automatically at every screen change).

What is SAP LUW or Update Transaction?

Update transaction (or “SAP LUW”)

This is a set of updates terminated by an ABAP/4 commit. A SAP LUW may last much longer than a database LUW, since most update processing extends over multiple transaction screens. The programmer terminates an update transaction by issuing a COMMIT WORK statement.

What happens if only one of the commands SET SCREEN and LEAVE SCREEN is used without using the other?

If we use SET SCREEN without LEAVE SCREEN, the program finishes processing for the current screen before branching to <scr no>. If we use LEAVE SCREEN without a SET SCREEN before it, the current screen process will be terminated and branch directly to the screen specified as the default next-screen in the screen attributes.

What is the significance of the screen number ‘0’?

In “calling mode”, the special screen number 0 (LEAVE TO SCREEN 0) causes the system to jump back to the previous call level. That is, if you have called a screen sequence with CALL SCREEN leaving to screen 0 terminates the sequence and returns to the calling screen. If you have not called a screen sequence, LEAVE TO SCREEN 0 terminates the transaction.

What does the ‘SUPPRESS DIALOG’ do?

Suppressing of entire screens is possible with this command. This command allows us to perform screen processing “in the background”. Suppressing screens is useful when we are branching to list-mode from a transaction dialog step.

What is the significance of the memory table ‘SCREEN’?

At runtime, attributes for each screen field are stored in the memory table called ‘SCREEN’. We need not declare this table in our program. The system maintains the table for us internally and updates it with every screen change.

What are the fields in the memory table ‘SCREEN’?

Name	Length	Description
NAME	30	Name of the screen field
GROUP1	3	Field belongs to field group 1
GROUP2	3	Field belongs to field group 2
GROUP3	3	Field belongs to field group 3
GROUP4	3	Field belongs to field group4
ACTIVE	1	Field is visible and ready for input.
REQUIRED	1	Field input is mandatory.
INPUT	1	Field is ready for input.
OUTPUT	1	Field is display only.
INTENSIFIED	1	Field is highlighted
INVISIBLE	1	Field is suppressed.
LENGTH	1	Field output length is reduced.
DISPLAY_3D	1	Field is displayed with 3D frames.
VALUE_HELP	1	Field is displayed with value help.

Why grouping of fields is required? What is the max no of modification groups for each field?

If the same attribute need to be changed for several fields at the same time these fields can be grouped together.

We can specify up to four modification groups for each field.

What are the attributes of a field that can be activated or deactivated during runtime?

Input, Output, Mandatory, Active, Highlighted, Invisible.

What is a screen group? How it is useful?

Screen group is a field in the Screen Attributes of a screen. Here we can define a string of up to four characters which is available at the screen runtime in the SY-DNGR field. Rather than maintaining field selection separately for each screen of a program, we can combine logically associated screens together in a screen group.

What is a Subscreen? How can we use a Subscreen?

A subscreen is an independent screen that is displayed in a n area of another (“main”) screen. To use a subscreen we must call it in the flow logic (both PBO and PAI) of the main screen. The CALL SUBSCREEN statement tells the system to execute the PBO and PAI events for the subscreen as part of the PBO or PAI events of the main screen. The flow logic of your main program should look as follows:

PROCESS BEFORE OUTPUT.

CALL SUBSCREEN <area> INCLUDING ‘<program>’ ‘<screen>’.

PROCESS AFTER INPUT.

CALL SUBSCREEN <area>.

Area is the name of the subscreen area you defined in your main screen. This name can have up to ten characters. Program is the name of the program to which the subscreen belongs and screen is the subscreen’s number.

What are the restrictions on Subscreens?

Subscreens have several restrictions. They cannot:

- Set their own GUI status
- Have a named OK code
- Call another screen
- Contain an AT EXIT-COMMAND module
- Support positioning of the cursor.

How can we use / display table in a screen?

ABAP/4 offers two mechanisms for displaying and using table data in a screen. These mechanisms are TABLE CONTROLS and STEP LOOPS.

What are the differences between TABLE CONTROLS and STEP LOOPS?

TABLE CONTROLS are simply enhanced STEP LOOPS that display with the look and feel of a table widget in a desktop application. But from a programming standpoint, TABLE CONTROLS and STEP LOOPS are almost exactly the same. One major difference between STEP LOOPS and TABLE CONTROLS is in STEP LOOPS their table rows can span more than one time on the screen. By contrast the rows in a TABLE CONTROLS are always single lines, but can be very long. (Table control rows are scrollable). The structure of table control is different from step loops. A step loop, as a screen object, is simply a series of field rows that appear as a repeating block. A table control, as a screen object consists of: i) table fields (displayed in the screen) ii) a control structure that governs the table display and what the user can do with it.

What are the dynapro keywords?

FIELD, MODULE, SELECT, VALUES and CHAIN are the dynapro keywords.

Why do we need to code a LOOP statement in both the PBO and PAI events for each table in the screen?

We need to code a LOOP statement in both PBO and PAI events for each table in the screen. This is because the LOOP statement causes the screen fields to be copied back and forth between the ABAP/4 program and the screen field. For this reason, at least an empty LOOP....ENDLOOP must be there.

The field SY-STEPL refers to the index of the screen table row that is currently being processed. The system variable SY-stepl only has a meaning within the confines of LOOP....ENDLOOP processing. Outside the loop, it has no valid value.

How can we declare a table control in the ABAP/4 program?

Using the syntax controls <table control name> type tableview using screen <scr no>.

Differentiate between static and dynamic step loops.

Step loops fall into two classes: Static and Dynamic. Static step loops have a fixed size that cannot be changed at runtime. Dynamic step loops are variable in size. If the user re-sizes the window the system automatically increases or decreases the number of step loop blocks displayed. In any given screen you can define any number of static step loops but only a single dynamic one.

What are the two ways of producing a list within a transaction?

By submitting a separate report.

By using leave to list-processing.

What is the use of the statement Leave to List-processing?

Leave to List-processing statement is used to produce a list from a module pool. Leave to list processing statement allows to switch from dialog-mode to list-mode within a dialog program.

When will the current screen processing terminates?

A current screen processing terminates when control reaches either a Leave-screen or the end of PAI.

How is the command Suppress-Dialog useful?

Suppressing entire screens is possible using this command. This command allows us to perform screen processing “in the background”. The system carries out all PBO and PAI logic, but does not display the screen to the user. Suppressing screens is useful when we are branching to list-mode from a transaction dialog step.

What happens if we use Leave to list-processing without using Suppress-Dialog?

If we don’t use Suppress-Dialog to next screen will be displayed but as empty, when the user presses ENTER, the standard list output is displayed.

How the transaction that are programmed by the user can be protected?

By implementing an authority check.

What are the modes in which any update tasks work?

Synchronous and Asynchronous.

What is the difference between Synchronous and Asynchronous updates?

A program asks the system to perform a certain task, and then either waits or doesn’t wait for the task to finish. In synchronous processing, the program waits: control returns to the program only when the task has been completed. In asynchronous processing, the program does not wait: the system returns control after merely logging the request for execution.

SAP system configuration includes Dialog tasks and Update tasks.

Dialog-task updates are Synchronous updates.

Update-task updates are Asynchronous updates.

What is the difference between Commit-work and Rollback-Work tasks?

Commit-Work statement “performs” many functions relevant to synchronized execution of tasks. Rollback-work statement “cancels: all reuests relevant to synchronized execution of tasks.

What are the different database integrities?

- Semantic Integrity.
- Relational Integrity.
- Primary Key Integrity.
- Value Set Integrity.
- Foreign Key integrity and
- Operational integrity.

All SAP Databases are Relational Databases.

What is SAP locking?

It is a mechanism for defining and applying logical locks to database objects.

What does a lock object involve?

The tables.

The lock argument.

What are the different kinds of lock modes?

Shared lock

Exclusive lock.

Extended exclusive list.

How can a lock object be called in the transaction?

By calling Enqueue<lock object> and Dequeue<lock object> in the transaction.

What are the events by which we can program “help texts” and display “possible value lists”?

-PROCESS ON HELP-REQUEST (POH).

-PROCESS ON VALUE-REQUEST (POV).

In what ways we can get the context sensitive F1 help on a field?

- Data element documentation.
- Data element additional text in screen painter.
- Using the process on help request event.

What is roll area?

A roll area contains the program's runtime context. In addition to the runtime stack and other structures, all local variables and any data known to the program are stored here.

How does the system handle roll areas for external program components?

- Transactions run in their own roll areas.
- Reports run in their own roll areas.
- Dialog modules run in their own roll areas
- Function modules run in the roll area of their callers.

Does the external program run in the same SAP LUW as the caller, or in a separate one?

- Transactions run with a separate SAP LUW
- Reports run with a separate SAP LUW.
- Dialog modules run in the same SAP LUW as the caller
- Function modules run in the same SAP LUW as the caller.

The only exceptions to the above rules are function modules called with IN UPDATE TASK (V2 function only) or IN BACKGROUND TASK (ALE applications). These always run in their own (separate) update transactions.

What are function modules?

Function modules are general-purpose library routines that are available system-wide.

What are the types of parameters in the function modules?

In general, function module can have four types of parameters:

- EXPORTING: for passing data to the called function.
- IMPORTING: for receiving data returned from the function module.
- TABLES: for passing internal tables only, by reference (that is, by address).
- CHANGING: for passing parameters to and from the function.

What is the difference between Leave Transaction and Call Transaction?

In contrast to LEAVE TO TRANSACTION, the CALL TRANSACTION statement causes the system to start a new SAP LUW. This second SAP LUW runs parallel to the SAP LUW for the calling transaction.

How can we pass selection and parameter data to a report?

There are three options for passing selection and parameter data to the report.

- Using SUBMIT...WITH
- Using a report variant.
- Using a range table.

How to send a report to the printer instead of displaying it on the screen?

We can send a report to the printer instead of displaying it on the screen. To do this, use the keywords TO SAP-SPOOL:

SUBMIT RSFLFIND...TO SAP-SPOOL DESTINATION 'LT50'.

How can we send data to external programs?

Using SPA/GPA parameters(SAP memory).

Using EXPORT/IMPORT data (ABAP/4 memory)

What are SPA/GPA parameters (SAP memory)

SPA/GPA parameters are field values saved globally in memory. There are two ways to use SPA/GPA parameters:

By setting field attributes in the Screen Painter.

By using the SET PARAMETER or GET PARAMETER statements

1. The source code for a report is stored in database table **dd010s**.

2. The customer name range is two to eight characters long and the program name must start with the letter y or z.

3. You can convert SAP script to Smartforms using **SF_MIGRATE** standard report. Just give your SAP Script name, and then this Report automatically converts your SAP Script to Smartforms.

4. How can I get ASCII value of any letter? Is there any function?

This is how you can do it:

```
report demtest.  
data : c.  
field-symbols : <n> type x.  
data : rn type i.  
c = 'A'.  
assign c to <n> casting.  
move <n> to rn.  
write rn.
```

This will convert 'A' to 65.

*going from 66 to B

```
data : i type i value 66.  
data : x type x.  
field-symbols : <fc> type c.  
move i to x.  
assign x to <fc> casting type c.  
move <fc> to c.  
write c.
```

5. What are the 3 types of function modules in SAP?

- (i) Remote enabled
- (ii) Normal
- (iii) Updated

6. How many secondary indexes can we create in a table?

We can create 15 secondary indexes for a table.

7. Is main window mandatory in smartforms?

Nothing is mandatory for smartforms

8. Which one is better joins or views and why?

Innerjoin, because it gives one to one relation with db tables and u can get exact results

9. **JAWS is a screen reader software**, which reads out all the elements in the screen. This software is commonly used by blind people to work on the system. From the next release onwards, SAP is going to release a separate version for this purpose.

10. What is the typical structure of an ABAP program?

Modular Structure

11. What are field symbols and field groups.? Have you used "component idx of structure" clause with field groups?

Field-Symbol.

A field-symbol is a pointer you can dynamically assign to a field. After assignment, you use the field-symbol anywhere in your program in place of the actual field name. Use the field-symbol statements to define a field-symbol and use assign to assign a field to it. The field-symbol name must begin and end with angle brackets. A simple example.

Report ztest.

Data f1(3) value 'ABC'.

Field-symbols <f>.

Assign f1 to <f>. “<f> can now be used in place of f1.

Write :/ <f>. “write the contents of f1

<f> = 'XYZ'. “assigns a new value to f1.

Write :/ f1.

Field group.

A field group combines several existing fields together under one name. You use the INSERT statement to determine which fields belong to a field group at runtime.

Example

FIELD-GROUPS: HEADER, ORDER, PRODUCT.

Note:

Neither defining a field group (statically) using FIELD-GROUPS nor filling a field group (dynamically) with INSERT generates more memory. Rather, there exists for each field group element a pointer to an (existing) field.

You cannot define groups in methods.

12. What should be the approach for writing a BDC program?

1. Analyzing data from local file.
2. Analyzing transaction
3. Declaring internal table
 - First Internal table similar to structure like local file
 - Declaring internal table like BDCDATA
4. Transferring data from local file to internal table
5. Population of BDCDATA.

From Internal table the data is transferred to database table by two ways. I.e. Session method or Call transaction.

13. What is a batch input session?

In this method you transfer data from internal table to database table through sessions.

In this method an ABAP/4 program reads the external data that is to be entered in the SAP system and stores the data in a session. A session stores the actions that are required to enter your data using normal SAP transactions. I.e. Data is transferred to session, which in turn transfer data to database table.

Session is intermediate step between internal table and database table. Data along with its action is stored in session i.e. Data for screen fields to which screen it is passed, the program name behind it, and how next screen is processed.

When the program has finished generating the session, you can run the session to execute the SAP transactions in it. You can either explicitly start and monitor a session or have the session run in the background processing system.

Unless session is processed, the data is not transferred to database table.

14. What is the alternative to batch input session?

Call Transaction Method

15. A situation: An ABAP program creates a batch input session. We need to submit the program and the batch session in background. How to do it?

The transaction for background processing is SM36.

Defining background jobs.

It is two steps process

- You first define the job and then you have to release it.

A Job in background processing is a series of steps that can be scheduled and step is a program for background processing. Step involves the following

. Job Name
. Job Class
. Job Steps

16. What is the difference between a pool table and a transparent table and how they are stored at the database level?

A transparent table in dictionary has one-to-one relationship with a table in the database. For each transparent table definition in the dictionary, there is one associated table in the database. The database table has the same names as the R/3 table definition.

A pooled table in the R/3 has a many-to-one relationship with a table in the database. For one table in the database has a different name than the tables in the DDIC, it has a different number of fields and the fields have different names as well. Pooled tables are a SAP proprietary construct.

17. What are the problems in processing batch input sessions? How is batch input process different from processing online?

If batch-input session is terminated with errors, it appears in the list of INCORRECT session and it can be processed again. To correct session, first you can analyze the session. The analysis function allows to determine which screen and value produced error. If you find small errors in the data, you can correct them interactively, otherwise you need to modify batch input program, which are generated the session or many times even the data file.

While batch input is a two-step procedure, call transaction does both steps online one right after the other. In this method, you call a transaction from your program.

18. What do you define in the domain and data element?

Domain: Parameters to be passed

1. Data Type: where you need to enter the data type available in SAP
2. Field Length: Field Length in the number of valid position
3. Value Table: name of a table to be entered, the fields referring to the domain may only assume values contained in the value table.

Data Element: Parameters to be passed when creating a data element.

1. Short Text: Mandatory Fields
2. Domain: A mandatory field.
3. Text Element: You can enter description in short or long text for the field. This text is used when entering data for these fields.

19. What are the different types of data dictionary objects?

1. Tables
2. Structures
3. Views
4. Data Element
5. Domains
6. Lock Objects
7. Math code objects

20. How many types of tables exists and what are they in data dictionary?

1. Transparent Tables
2. Pooled Tables
3. Cluster Tables

Transparent tables do exists with the same structure both in data dictionary as well as in the database, exactly with same data and fields while other two are not transparent in the sense that they are not manageable directly using database system tools. You cannot use native SQL on these tables. Pool or cluster tables are logical tables, which are arranged as records of transparent tables.

21. What is the step-by-step process to create a table in data dictionary?

Steps to create a table

1. Create Domain
2. Create Data Element
3. Create actual table

22. Can a transparent table exist in data dictionary but not in the database physically?

No

23. What are the domains and data elements?

Domain: They are formal definition of data types from a technical point of view. They set attributes such as data type, length possible value range and so.

Data Element: They are definitions of the properties and type for a table field. It is an intermediate object between the object type domain and the table field. A field in R/3 system is always associated with a data element, which at the same time is related to domain.

24. Can you create a table with fields not referring to data elements?

Yes

25. What is the advantage of structures? How do you use them in the ABAP programs?

You can use structures to define identical work area in multiple programs. You can include a structure within another structures and tables.

Like a table, it can be used within a program on the tables' statement to define a work area.

26. What does an extract statement do in the ABAP program?

Writes all fields of the field group fg (FIELD-GROUPS) as an entry in a sequential dataset. If you have defined a field group HEADER, its fields precede each entry as a sort key. Afterwards, you can use SORT and LOOP... ENLOOP to sort or process the dataset respectively. No further EXTRACT statements are possible after this.

General:

1. As soon as you have extracted a dataset using EXTRACT, you can no longer extend the field group-using INSERT. In particular, you cannot change the HEADER field group at all after the first EXTRACT (regardless of the field group to which it applied).
2. Large extract datasets are not stored in main memory. Instead, they are kept in an external auxiliary file. You can set the directory in which this file is created using the SAP profile parameter DIR_EXTRACT. The default directory is the SAP data directory (SAP profile parameter DIR_DATA).

27. What is a collect statement? How is it different from append?

Collect searches in the internal table for an entry, all of whose alphanumeric fields are identical with those of the entry in the work area or header line. If such an entry is found, COLLECT adds all numeric fields from work area or header line to the corresponding fields in the table entry. Otherwise the collect statement appends the contents of the work area or header line to the end of the table.

The APPEND statement stores the contents of the header line at the end of the internal table.

28. What is open sql vs native sql?

Open SQL allow you to access database tables declared in the ABAP dictionary regardless of the database platform that you R/3 system is using. Native SQL allows you to use database specific SQL statements in an ABAP program. This means that you can use database tables that are not administered by the ABAP dictionary and therefore integrate data that is not part of the R/3 system.

As a rule, an ABAP program containing database-specific SQL statements will not run under different database system. If your program will be used on more than one database platform, only use Open SQL statements.

29. What does an EXEC SQL stmt do in ABAP? What is the disadvantage of using it?

Executes the Native SQL statements enclosed between EXEC SQL and ENDEXEC statements. Unlike Open SQL Native SQL statements can address tables that are not declared in the ABAP Dictionary.

Example

Creating the Table AVERI_CLNT:

```
EXEC SQL.
  CREATE TABLE AVERI_CLNT (
    CLIENT CHAR(3) NOT NULL,
    ARG1 CHAR(3) NOT NULL,
    ARG2 CHAR(3) NOT NULL,
    FUNCTION CHAR(10) NOT NULL,
    PRIMARY KEY (CLIENT, ARG1, ARG2)
  )
ENDEXEC.
```

In a Native SQL statement, data is passed between the ABAP program and the database using host variables. A host variable is an ABAP variable that is identified as such in the Native SQL statement by a preceding colon (:).

Example

Displaying an extract from the table AVERI_CLNT:

```
DATA: F1(3), F2(3), F3(3).
F3 = '1'.
EXEC SQL.
  SELECT CLIENT, ARG1 INTO :F1, :F2 FROM AVERI_CLNT
  WHERE ARG2 = :F3
ENDEXEC.
WRITE: / F1, F2.
```

To simplify the form of the INTO lists in the SELECT statement, you can, as in Open SQL, specify a single structure as the target area.

30. What is the meaning of ABAP editor integrated with ABAP data dictionary?

Integration means if you double-click an object to select it, the Workbench automatically launches the tool that was used to create the object.

SAP has developed the Object Navigator to help you to organize your application development in this integrated environment. It provides a context that makes it easier for you to trace the relationships between objects in a program. Rather than working with tools and recalling development objects, you work with objects and allow the Workbench to launch the appropriate tool for an object.

31. What are the events in ABAP language?

An ABAP/4 using events that are invoked by the user actions. Processing blocks are defined by event-keywords and are executed on invocation of certain relevant events.

By default, the event start-of-selection is attached to all events in ABAP/4. In your programs you can define a processing block and attach this block to an event keyword.

32. What is an interactive report? What is the obvious difference of such report compared with classical type reports?

A classical report consists of one program that creates a single list. This means that when the list is displayed, it has to contain all data requested, regardless of the number of details the user wants to see.

Interactive reporting allows the user to participate in retrieving and presenting data at each level during the session. Instead of presenting one extensive and detailed list with classified information. With interactive reporting you can create a condensed basic list from which the user can call detailed information by positioning the cursor and entering commands.

Detailed information is presented in secondary list. A secondary list may either overlay the basic list completely or appear in an additional dialog window on the same screen. The secondary list can itself be interactive again. The basic list is not deleted when secondary list is created.

User can interact with system by

- double clicking or pressing f2
- selecting menu option

Like classical report the interactive report is also event driven. Both the action mentioned above trigger events and code is written to handle these events. The events triggered by this action are as follows

- At line-selection
- At user-command

Interactive report consists of one basic list and 20 secondary list. Basic list is produced by START-OF-SELECTION event. When the user double clicks on the basic list or chooses the menu option, the secondary list is produced. All the events associated with classical report except end-of-page are applicable only to basic list.

33. What is a drill down report?

Interactive reports are called drill down report.

34. How do you write a function module in SAP? Describe.

Function modules are procedures that are defined in function groups (special ABAP programs with type F) and can be called from any ABAP program. Function groups act as containers for function modules that logically belong together. You create function groups and function modules in the ABAP Workbench using the Function Builder.

Function modules allow you to encapsulate and reuse global functions in the R/3 System. They are stored in a central library. The R/3 System contains a wide range of predefined function modules that you can call from any ABAP program. Function modules also play an important role in database updates and in remote communication between R/3 Systems or between an R/3 System and a non-SAP system.

Unlike subroutines, you do not define function modules in the source code of your program. Instead, you use the Function Builder. The actual ABAP interface definition remains hidden from the programmer. You can define the input parameters of a function module as optional. You can also assign default values to them. Function modules also support exception handling. This allows you to catch certain errors while the function module is running. You can test function modules without having to include them in a program using the Function Builder.

The Function Builder also has a release process for function modules. This ensures that incompatible changes cannot be made to any function modules that have already been released. This applies particularly to the interface. Programs that use a released function module will not cease to work if the function module is changed.

Function modules perform tasks of general interest to other programmers. Usually these tasks are well-defined functions that all users need, regardless of application. Some well-defined tasks include performing tax calculations, determining factory calendar dates, and calling frequently used dialogs.

When you write ABAP routines that other programmers might use, you should define these routines as function modules. This means that you develop them in the Function Builder as follows:

1. Check whether a suitable function module already exists. If not, proceed to step 2.
2. Create a function group, if no appropriate group exists yet.
3. Create the function module.
4. Define the function module interface by entering its parameters and exceptions.
5. Write the actual ABAP code for the function module, adding any relevant global data to the TOP include.
6. Activate the module.
7. Test the module.
8. Document the module and its parameters for other users.
9. Release the module for general use.

Runtime Considerations

There are some runtime considerations you should be familiar with when writing function modules:

- The CALL FUNCTION statement can pass import, export, and changing parameters either by value or by reference. Table parameters are always transferred by reference.
- If you declare the parameters with reference to ABAP Dictionary fields or structures, the system checks the type and length when the parameters are transferred. If the parameters from the calling program do not pass this check, the calling program terminates.
- At runtime, all function modules belonging to a function group are loaded with the calling program. As a result, you should plan carefully which functions really belong in a group and which do not. Otherwise, calling your function modules will unnecessarily increase the amount of memory required by the user.

35. What are the exceptions in function module?

Our function module needs an exception that it can trigger if there are no entries in table those meets the selection criterion. For example the exception NOT_FOUND serves this function.

36. What is a function group?

Function groups are containers for function modules. You cannot execute a function group. When you call a function module, the system loads the whole of its function group into the internal session of the calling program (if it has not already been loaded).

The name of a function group can be up to 26 characters long. This is used by the system to create the components of the group (main program and corresponding include programs). When you create a function group or function module in the function builder, the main program and include programs are generated automatically.

37. How are the date and time field values stored in SAP?

1. Date is stored in SY-DATUM.
2. Time is stored in SY-UZEIT.

38. What are the fields in a BDC_Tab Table?

BDCTAB is like BDCDATA. The fields are...

1. Program: Name of module pool program associated with the screen set this field only for the first record for the screen.
2. Dynpro: Number of the screen. Length(4). Set this field only in the first record for the screen.
3. Dynbegin: Indicates the first record for the screen. Length(1). Set this period to 'X' only for the first record for the screen. (Reset to ' ' blank for all other records).
4. Fnam: Name of a field in the screen. Length(35). The Fnam field is not case-sensitive.
5. Fval: Value for the field named in Fnam. Length(132). The fval field is case-sensitive. Values assigned to this field are always padded on the right if they are less than 132 characters. Values must be in character format.

39. Name a few data dictionary objects?

1. Tables
2. Structures
3. Views
4. Data Element
5. Domains
6. Lock Objects
7. Math code objects

40. What happens when a table is activated in DD?

When tables, types (data elements, structures, table types) and views are activated, they are placed at the disposal of the runtime environment in the form of runtime objects. These runtime objects contain the information about the object in a form that is optimal for access by ABAP programs and screens. The runtime objects are buffered so that ABAP programs and screens can access the information relevant to them quickly.

41. What is a check table and what is a value table?

Check table specified in the foreign key for the field. A foreign key links two tables T1 and T2 by assigning fields of table T1 to the primary key fields of table T2.

The T1 is called foreign key table (dependent table) and table T2 the check table (referenced table).

In some cases you can see when you define a domain that all the table fields or structure components referring to this domain should be checked against a certain table. This information can be stored in the domain by entering a value table.

The system proposes the value table as check table when you try to define a foreign key for the field or component. This proposal can be overridden.

Example:

Domain S_CARR_ID (data type CHAR, length 3) in the flight Model describes the three-place code of the airlines. All the airlines are listed together with their codes in table SCARR. It is generally advisable to check fields referring to domain S_CARR_ID against table SCARR. SCARR is therefore entered as value table for domain S_CARR_ID. If you want to define a foreign key for a field referring to S_CARR_ID, SCARR is proposed as the check table.

A check is not implemented by simply entering a value table! The check against the value table only takes effect when a foreign key has been defined.

42. What are match codes? Describe?

A matchcode is a means of finding data records stored in the system. The matchcode is defined in the ABAP

Dictionary in two steps:

- You first define the relevant tables and fields for the search in a matchcode object. A matchcode object describes the set of all possible search paths for a search string.
- You then create one or more matchcode IDs for a matchcode object. A matchcode ID describes a special search path for a search string. The matchcode ID defines the fields or field combinations to be used in the search.

A material number must be entered in a screen field. Since the user cannot be expected to know this number, it must be possible to search for this number using the attributes of the corresponding material.

Several search paths are possible for this search. For example, you can search for the material number with the material name, the material class or the material manufacturer.

The corresponding match code object then comprises the fields for the material number, material name, material class and manufacturer. One match code ID corresponds to each search path. For example, ID A could describe the search for the material number by manufacturer. This ID only contains the fields for the material number and manufacturer.

The tables relevant for the search are included in a match code object. The table selection is based on one primary table. Further secondary tables can also be included, which are linked with the primary table by foreign keys. The fields of the match code object can then be selected from the base tables.

A match code object is not stored physically. It only describes a complete logical view on one or more tables.

43. What transactions do you use for data analysis?

ST05

44. What is table maintenance generator?

SE55

The Generate table maintenance dialog component creates standardized maintenance dialogs for tables and views. These dialogs can also be used to maintain table or view contents.

Integration

The component provides a standardized maintenance interface for many customizing activities. It is also useful as a customer table or view input tool.

Table or view maintenance dialogs are created in the ABAP/4 Workbench under the menu path *Development* ® *Other tools* ® *Gen.tab.maint.dialog*. To maintain table or view contents choose *Services* ® *Ext. tab.maint.* at any time.

Maintenance dialogs and data which were created, changed or deleted with the maintenance dialog can be transported into other R/3 Systems.

Functionality

The component creates maintenance dialogs which are standardized in their:

- functionality
- interface
- maintenance screen
- navigation
- enhancement options
- maintenance

44. What are ranges? What are number ranges?

You can use the RANGES statement to create internal tables of the same type as selection tables.

RANGES <rangetab> FOR <f>.

This statement is simply a shortened form of the following statements:

```
DATA: BEGIN OF <rangetab> OCCURS 0,  
      SIGN(1),  
      OPTION(2)  
      LOW LIKE <f>,  
      HIGH LIKE <f>,  
      END OF <rangetab>.
```

Internal tables created with RANGES have the same structure as selection tables, but they do not have the same functionality.

Selection tables created with RANGES are not components of the selection screen. As a result, no relevant input fields are generated. Also, you cannot use a RANGES table as a data interface in program <prog> called by the following statement:

```
SUBMIT <prog> WITH <rangetab> IN <table>.
```

However, you can use RANGES to create the table <table> in the calling program. The main function of RANGES tables is to pass data to the actual selection tables without displaying the selection screen when executable programs are called.

Although you can use RANGES tables like actual selection tables in the WHERE clause of Open SQL statements and in combination with the IN operator in logical expressions, they are not linked to a database table. This means that RANGES tables:

- are not passed like selection criteria to logical databases.
- cannot be used with the shortened form of selection tables in logical expressions.

- cannot be used like selection criteria in GET events
REPORT DEMO1.
RANGES S_CARRID FOR SPFLI-CARRID.
S_CARRID-SIGN = 'I'.
S_CARRID-OPTION = 'EQ'.
S_CARRID-LOW = 'LH'.

APPEND S_CARRID.

SUBMIT DEMO2 WITH CARRID IN S_CARRID.

In this example, RANGES table S_CARRID is created with reference to column CARRID of database table SPFLI. Fields S_CARRID-LOW and S_CARRID-HIGH have the same type as CARRID. The header line of internal table S_CARRID is filled and appended to the table. Program DEMO2 is called. If DEMO2 is linked to logical database F1S, its selection screen contains the fields of selection criterion CARRID from the logical database. These fields are filled with the contents of the RANGES table.

Number Ranges

You can specify the number manually or it can be determined by the system from a pre-defined area (number range).

Example:

The relevant area of the Accounting document numbers in each company code per document type.

45. What are select options and what is the diff from parameters?

Select options you specify are displayed on the selection screen for the user to enter values.

In case of select option user can enter a range of values. But in parameters user can enter only single value.

46. How do you validate the selection criteria of a report? And how do you display initial values in a selection screen?

You can validate selection criteria of a report using at selection-screen event. You can display the initial values using default option of select-options.

47. What are selection texts?

Description of specified selection text (SELECT-OPTION, PARAMETER). This description appears on the selection screen.

Texts on the selection screen are stored as language-specific selection-texts in the program text elements.

48. What is CTS and what do you know about it?

A change request is a list in the system, where mainly contains the object to be transported. It also contains the transport type, the request category and the target system.

When the change request is created either manually or automatically the system assigns a number to it automatically and this number is known as change request number.

The change request records all modifications made to development object.

When the changes have been made and the change tasks have been released, the change request can be released.

SE09 will display and check all the change request.

49. When a program is created and need to be transported to production does selection texts always go with it? if not how do you make sure? Can you change the CTS entries? How do you do it?

Whenever selection texts are changed or created it will ask for change request, if a development class is assigned to the program. Using the change request number you can transport the selection texts.

Yes. You can change the CTS entries using transaction se09.

50. What is the client concept in SAP? What is the meaning of client Independent?

One of the most used client/server configurations with R/3 system is the tiered architecture, which separates a system's computer into 3 functional group.

Database server

Application Server

Presentation Server

Communication among the 3 tiers is accomplished by standard protocol servers like TCP/IP or CPIC (Common Programming Interface Communication).

Client Independence, which means that it is independent of all clients across the SAP system.

Difference between a check table and a value table

Value Table

This is maintained at Domain Level. When ever you create a domain , you can entered allowed values. For example you go to Domain SHKZG - Debit/credit indicator. Here only allowed values is H or S.

When ever you use this Domain, the system will forces you to enter only these values.

This is a sort of master check . To be maintained as a customization object. This mean that if you want to enter values to this table you have to create a development request & transport the same.

Check table

For example you have Employee master table & Employee Transaction table.

When ever an employee Transacts we need to check whether that employee exists , so we can refer to the employee master table.

This is nothing but a Parent & Child relationship . Here data can be maintained at client level , no development involved.

As per DBMS what we call foregin key table, is called as check table in SAP.

What is use of using HASHED TABLE?

Hashed table is useful when your have to work with very big internal table and to read it with "READ TABLE WITH KEY ..."

The time access is constant !

Definition of a Hashed Table:
"Defines the table as one that is managed with an internal hash procedure. You can imagine a hashed table as a set, whose elements you can address using their unique key. Unlike standard and sorted tables, you cannot access hash tables using an index. All entries in the table must have a unique key.

Access time using the key is constant, regardless of the number of table entries.

You can only access a hashed table using the generic key operations or other generic operations (SORT, LOOP, and so on). Explicit or implicit index operations (such as LOOP ... FROM to INSERT itab within a LOOP) are not allowed."

As long as your records has unique key(s), using hash table will give you a huge performance gain when dealing with large dataset. assuming in your case, 10000 record , and if the key is unique, use hash table. The main use of hash tables is for looking up fixed information from a key. So if you have a report that has personnel number and you want to display their name, you could use a hash table.

Thus:
Code:

```
types: begin of typ_pernr,  
       pernr like pa0001-pernr,  
       ename like pa0001-ename,  
       end of typ_pernr.  
data: ls_pernr type typ_pernr,
```

```

lt_pernr type hashed table of typ_pernr with unique key pernr.
...
select pernr ename into table lt_pernr from pa0001.
...
loop at itab.
  read table lt_pernr with table key pernr = itab-pernr
  into ls_pernr.
  write: ls_pernr-ename, itab-data.
endloop.

```

The Different Types of SAP Tables

Could anyone tell me what is the major difference between Standard tables, Pooled tables and Clusterd Tables.

A transparent table is a table that stores data directly. You can read these tables directly on the database from outside SAP with for instance an SQL statement.

Transparent table is a one to one relation table i.e. when you create one transparent table then exactly same table will create in data base and if is basically used to store transaction data.

A clustered and a pooled table cannot be read from outside SAP because certain data are clustered and pooled in one field.

One of the possible reasons is for instance that their content can be variable in length and build up. Database manipulations in Abap are limited as well.

But pool and cluster table is a many to one relationship table. This means many pool table store in a database table which is know as table pool.

All the pool table stored table in table pool does not need to have any foreign key relationship but in the case of cluster table it is must. And pool and cluster table is basically use to store application data.

Table pool can contain 10 to 1000 small pool table which has 10 to 100 records. But cluster table can contain very big but few (1 to 10) cluster table.

For pool and cluster table you can create secondary index and you can use select distinct, group for pool and cluster table. You can use native SQL statement for pool and cluster table.

A structure is a table without data. It is only filled by program logic at the moment it is needed starting from tables.

A view is a way of looking at the contents of tables. It only contains the combination of the tables at the basis and the way the data needs to be represented. You actually call directly upon the underlying tables.

Fields of Internal Tables

SY-TABIX Current line of an internal table. SY-TABIX is set by the statements below, but only for index tables. The field is either not set or is set to 0 for hashed tables.

APPEND sets SY-TABIX to the index of the last line of the table, that is, it contains the overall number of entries in the table.

COLLECT sets SY-TABIX to the index of the existing or inserted line in the table. If the table has the type HASHED TABLE, SY-TABIX is set to 0.

LOOP AT sets SY-TABIX to the index of the current line at the beginning of each loop pass. At the end of the loop, SY-TABIX is reset to the value that it had before entering the loop. It is set to 0 if the table has the type HASHED TABLE.

READ TABLE sets SY-TABIX to the index of the table line read. If you use a binary search, and the system does not find a line, SY-TABIX contains the total number of lines, or one more than the total number of lines. SY-INDEX is undefined if a linear search fails to return an entry.

SEARCH <itab> FOR sets SY-TABIX to the index of the table line in which the search string is found.
SY-TFILL :After the statements DESCRIBE TABLE, LOOP AT, and READ TABLE, SY-TFILL contains the number of lines in the relevant internal table.
SY-TLENG :After the statements DESCRIBE TABLE, LOOP AT, and READ TABLE, SY-TLENG contains the length of the lines in the relevant internal table.
SY-TOCCU :After the statements DESCRIBE TABLE, LOOP AT, and READ TABLE, SY-TLENG :contains the initial amount of memory allocated to the relevant internal table.

Difference between extract and collect statements

What is the difference between 'extract' and 'collect' statements?

Once you have declared the possible record types as field groups and defined their structure, you can fill the extract dataset using the following statements:

EXTRACT <fg>.

When the first EXTRACT statement occurs in a program, the system creates the extract dataset and adds the first extract record to it. In each subsequent EXTRACT statement, the new extract record is added to the dataset.

Each extract record contains exactly those fields that are contained in the field group <fg>, plus the fields of the field group HEADER (if one exists). The fields from HEADER occur as a sort key at the beginning of the record. If you do not explicitly specify a field group <fg>, the EXTRACT statement is a shortened form of the statement extracts used in field groups (version 2.x - obsolete these days as noone uses field groups anymore), and collect is used to accumulate the contents of a field if X no. of keys are the same.

EXTRACT HEADER.

When you extract the data, the record is filled with the current values of the corresponding fields. As soon as the system has processed the first EXTRACT statement for a field group <fg>, the structure of the corresponding extract record in the extract dataset is fixed. You can no longer insert new fields into the field groups <fg> and HEADER. If you try to modify one of the field groups afterwards and use it in another EXTRACT statement, a runtime error occurs.

By processing EXTRACT statements several times using different field groups, you fill the extract dataset with records of different length and structure. Since you can modify field groups dynamically up to their first usage in an EXTRACT statement, extract datasets provide the advantage that you need not determine the structure at the beginning of the program.

Collect:

When the line is inserted, the system checks whether there is already a table entry that matches the key. If there is no corresponding entry already in the table, the COLLECT statement has the same effect as inserting the new line. If an entry with the same key already exists, the COLLECT statement does not append a new line, but adds the contents of the numeric fields in the work area to the contents of the numeric fields in the existing entry.

You should only use the COLLECT statement if you want to create summarized tables. If you use other statements to insert table entries, you may end up with duplicate entries.

- Logging on without being authorized
Client 066 usually exists in a SAP system because of EarlyWatch services. Often this client does not have master users. If it is true, anyone can log into the system using the client 066, user SAP*, and password PASS. Enjoy yourself.
- Long messages on footer
Click on the message and hold the mouse button. After moving the mouse to the left side.
- Direct input logs
The transaction BMV0 (direct input logs) shows all direct input logs.
- Filling up an empty date field quickly
Strike the key F4 (or click on matchcode symbol) and press ESCAPE. The current date is automatically set.
- Setting up module FI/CO without using IMG
Almost all parameters can be set using the transactions ORFA (Asset Accounting), ORFB (Financial Accounting), and ORKS (Cost Center Accounting).
- Deleting cost elements and cost centers
Since they have no postings you can use the transaction KA04 for deleting cost elements and KS04 for deleting cost centers.
- Displaying check object when not authorized
Soon after the lock try to access the transaction SU53. It reports the last objects verified and also the respective values.
- Table analyses between two systems
The contents of a table between two systems can be checked through the transaction OY19.
- Correction and transport system
The transaction SE10 provides the easiest way to manage any request/transport and corrections.
- General command field formats

/n Skip to the next record if you are processing one batch input session
 /bend Cancel a batch input foreground process
 /nend Close all R/3 sessions and logoff
 /nxxxx Call the transaction xxxx in the same session
 /o Generate a session list
 /oxxxx Call the transaction xxxx in an additional session
 /i Delete the current session
 /h Turn the debug mode on
 /\$tab Reset all buffers (for System Administrators)
 /\$sync Synchronize instances buffers (for System Administrators)

- Report command field formats

%pri Print the current report
 %pc Download the current report
 %sc Call the find function
 p+ Go to the next page
 p- Go to the previous page
 p++ Go to the last page
 p-- Go to the first page

- Helpful reports

RSCLTCOP	Copy tables across clients
RSVAVGL00	Table adjustment across clients
RSINCL00	Extended program list
RSBDCSUB	Release batch-input sessions automatically
RSTXSCR	Transport SAPscript files across systems
RSORAREL	Get the Oracle Release
RGUGBR00	Substitution/Validation utility
RSPARAM	Display all instance parameters
RSUSR003	Check the passwords of users SAP* and DDIC in all clients
RSUSR006	List users last login

- Meaning of info structures' first letter

A	Pricing
B	Output determination
C	Account determination
D	Material determination
E	Rebates
F	Index
G	Listing and Exclusion
H	Batch determination
I	Profile determination
S	Statistics
X	Statistics extra

- Unconditional mode when importing or exporting a request/transport
Run the command *R3trans -u* under user «SysID»adm.

- Reapplying hot packages
If you accidentally applied hot packages out of sequence for instance. Use the transaction SM31 to modify table PAT03. You have to choose the desired patch and click on delete entry.

- Main return codes of *tp* program

0	Successfully done
4	Warnings occurred
8	Errors occurred
12	Fatal errors occurred
16	Internal errors occurred

- Scheduling of system maintenance jobs

RSBTCDEL	Clean the old background job records
RSDBCRCO	Clean batch input session log
RSPO0041	Removing old spooling objects
RSSNAPDL	Clean the old ABAP error dumps

- List of most used SAP extensions and their components

CUST1	MENUS000+C01	Customer option in the Office menu
CUST2	MENUS000+C02	Customer option in the Logistics menu
CUST3	MENUS000+C03	Customer option in the Accounting menu
CUST4	MENUS000+C04	Customer option in the Human Resources menu
CUST5	MENUS000+C05	Customer option in the Information Systems menu
CUST6	MENUS000+C06	Customer option in the Tools menu
CUST7	MENUS000+C07	Customer option in the System menu
ZXUSR001	Exit_saplsusf_001	At login time
SAPMF02D	Exit_sapmf02d_001	When saving customer master data

SAPMF02K	Exit_sapmf02k_001	When saving vendor master data
M61X0001	Exit_saplm61c_001	When processing MRP planning
M61X0001	Exit_sapmm61x_001	When processing MRP planning
FYTX0001	Exit_saplv61a_001	Modifications in pricing procedures
MBCF0002	Exit_sapmm07m_001	Checks for materials documents
SDVFX002	Exit_saplv60b_002	Link between SD and FI documents
M06B0003	Exit_sapmm06b_001	When saving MM documents

- Before going live
It is highly advisable to increase the next extend's size of some tables and their indexes even before initial loadings

FI	BKPF, BSEG, BSIS, BSAD, BSAK, BSID and BSIK
CO	COEJ, COEP, COKS, COSS and T811*
AM	ANL*
MM	MKPF, MSEG and BSIM
SD	VBAP, VBAK, VBEP, VBPA, LIKP, LIPS, VBRK, VBRP, VBKD, VBUK, VBUP and VBSS
PP	RESB and MDTB
Accross module	ATAB, TST03, TSP01, MCSI, KNVP, ACCTIT, COEP, APQD, RFBLG, CDCLS, SDBAD and from S000 to S999

- Locking the whole system
Using the command *tp locksyes «SysID»* only the user SAP* will be allowed to login. The command *tp unlocksyes «SysID»* cancels the lock.
- Connection between SAP R/3 and operating system
The command *sapevt* can be used to trigger an event from the operation system. Thus, a job previously defined within R/3 will be released.
- SQL code help
Run the command *oerr ora «error number»* under user ora«SysID».
- Oracle import and export explanations
Run the command *imp help=yes* under user ora«SysID». This format can also be used with *exp*, *impst*, and *expst*.

Note: Research based on version 3.0f under Unix, Oracle data base and Windows.

Some special features allow you wide modifications without changing standard SAP R/3 objects. Those techniques are not as widespread as they should be. In fact they are powerful tools.

- Field exit
After entering a value in a field, it can be checked through a field exit. The system makes the field value available to be checked and changed in an ABAP/4 function.
- User exit
Points previously set in the system that let you evaluate data. The fields available are also previously defined by SAP. All fields value available can be checked in an ABAP/4 program.
- Validation
It allows solid data entry regarding special rules. According to previous rules, the system can evaluate an entry and a message can appear on the user's terminal if a check statement is not met. A validation step contains prerequisite statement and check statement. Both of them are defined using Boolean Logic or calling an ABAP/4 form.
- Substitution
Fields contents can be changed using substitution. When data are being entered, the data can be substituted by another value regarding rules previously defined. A substitution step contains prerequisite statement, substitution value and substitution exit. All of them are defined using Boolean Logic or calling an ABAP/4 form.

- **Set**
Values or ranges of values are specified under a set name. Sets are easier to create and maintain instead of using tables. They give you more flexibility when maintaining your system.
- **Key**
It allows changes on field description according to data element. The short key word used on most screen to identify the corresponding field contents can be changed too.

Requirements & formulas
ABAP/4 forms that can be used to handle pricing procedures, rounding rules, copy and data transport Sales activities.

